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A continuum-mechanical theory of redundancy in elastostatic structures

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Dedicated to our teacher, doctoral advisor and academic role model Ekkehard Ramm on occasion of his 80th birthday

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ABSTRACT

In the present paper, theoretical foundations of redundancy in spatially continuous, elastostatic, and linear representations of structures are derived. Adopting an operator-theoretical perspective, the redundancy operator is introduced, inspired by the concept of redundancy matrices, previously described for spatially discrete representations of structures. Studying symmetry, trace, rank, and spectral properties of this operator as well as revealing the relation to the concept of statical indeterminacy, a continuum-mechanical theory of redundancy is proposed. Here, the notion "continuum-mechanical" refers to the representation being spatially continuous. Apart from the theory itself, the novel outcome is a clear concept providing information on the distribution of statical indeterminacy in space and with respect to different load carrying mechanisms. The theoretical findings are confirmed and illustrated within exemplary rod, plane beam, and plane frame structures. The additional insight into the load carrying behavior may be valuable in numerous applications, including robust design of structures, quantification of imperfection sensitivity, assessment of adaptability, as well as actuator placement and optimized control in adaptive structures.

1. Introduction

1.1. Motivation and goals

Exploring redundancies in a load carrying structure plays a crucial role in many aspects of its design and analysis. It may address safety, reliability and ductility in case of (partial) structural failure (e.g. Frangopol and Curley, 1987; Pandey and Barai, 1997), as well as the load carrying behavior. All these aspects are related to the distribution of internal constraint and thus to the ability of statically indeterminate structures to carry loads along different paths. The degree of statical indeterminacy is a system-inherent property of the spatially discrete, elastostatic representation of a truss or beam structure, presuming linear theory. It is defined as the difference between the number of unknown force quantities and the number of linearly independent equilibrium equations. Methods for its determination belong to the standard portfolio in structural mechanics. Although further specifications like e.g. decomposition into an internal and external part or determination with respect to special load directions are established, nowadays the property is mainly considered as one aggregated integer number, characterizing the entire structure (or entire substructures) without quantified information about its distribution in space and with respect to load carrying mechanisms.

The redundancy concept introduced in Bahndorf (1991), Ströbel (1995) and von Scheven et al. (2021), provides information on the distribution of statical indeterminacy in discrete representations of truss and beam structures given by the main-diagonal components of the redundancy matrix. This additional insight into the load carrying behavior may be valuable in numerous applications, including robust design of structures, quantification of imperfection sensitivity, assessment of adaptability as well as actuator placement and optimized control in adaptive structures. However, the findings published so far are mainly restricted to discrete representations and there is less scientific work available on structures in a spatially continuous setting. A theory on redundancy in continuous representations overcomes the restriction imposed by consideration of discrete representations and enhances insight into the redundancy concept.

Addressing this motivation, the present work forms a connection between concepts from applied mathematics and classical structural mechanics. Its purpose is to introduce theoretical foundations on redundancy embedded in a continuum-mechanical framework. The major novelty compared to previous work (especially compared to von Scheven et al., 2021) is the generalization of the concept of redundancy from discrete to continuous representations of structures. Fur this purpose, an operator-theoretical perspective is adopted and, amongst others, a redundancy operator is introduced. Studying symmetry, trace, rank, and spectral properties of this operator as well as revealing the relation to the concept of statical indeterminacy, a theory of redundancy is proposed. Apart from the theory itself, the novel outcome is a clear concept providing information on the distribution of statical indeterminacy in space and with respect to different load carrying mechanisms.

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The novel theoretical findings are confirmed and illustrated by several examples. This provides valuable knowledge in different kinds of applications, like e.g. in understanding theoretical opportunities and limits of adaptability of internal force quantities in adaptive structures. While the approach is general, the present work is limited to continuous, elastostatic, and linear representations of structures obeying statically determinate structural theories. The notion "statically determinate structural theory" denotes a theory with an equal number of internal force functions and equilibrium differential equations (e.g. Euler-Bernoulli beam theory). In these kinds of structures, redundancies stem from the support conditions and structural topology only.

1.2. State of the art

Several publications on structural redundancy embedded in a spatially discrete (or discretized) framework exist, but related research in the context of a continuum-mechanical framework is rare. Significant works within the former area are mentioned in the next two paragraphs, followed by relevant literature for the latter.

Driven by the analogy between Gaussian adjustment calculus in geodesy and in structural mechanics, which is well-known since more than a hundred years (von Scheven et al., 2021 for a selection of early publications), a concept of redundancy in statically indeterminate truss structures was developed in the group of Klaus Linkwitz in Stuttgart. Based on Linkwitz' work Linkwitz (1961), Bahndorf (1991) and Ströbel (1995) presented an idempotent matrix providing information on redundancy contributions and the spatial distribution of statical indeterminacy. Applying this approach, Eriksson and Tibert (2006), Ströbel and Singer (2008) investigated sensitivities due to manufacturing imperfections and Kou et al. (2017) proposed an index for structural robustness. In Tibert (2005) the approach by Bahndorf and Ströbel was extended to kinematically indeterminate structures by considering the concept of pseudo-inverse in structural computations (Pellegrino and Calladine, 1986; Pellegrino, 1990, 1993). Zhou et al. (2015) presented a method for the determination of distributed statical indeterminacy in kinematically indeterminate structures. According to Chen et al. (2018), this is identical to the formulation in Ströbel (1995) when the concept of pseudo-inverse is considered. In the context of reanalysis of discrete representations of structures, Kołakowski et al. (2008) used a so-called "strain influence matrix". The rank of this matrix was recognized to be equal to the degree of statical indeterminacy, however without describing the information on its spatial distribution.

Later, initiated by Ekkehard Ramm, the concept of redundancy distribution in discrete representations of structures was revisited and the redundancy matrix was introduced *"as a basis for additional insight into structural properties beyond the classical definition of statical indeterminacy"* (von Scheven et al., 2021). Based on their findings, applications of the redundancy concept in the design of adaptive structures were carried out recently (Wagner et al., 2018; Geiger et al., 2020). In the context of shape and stress control, a matrix identifiable as redundancy matrix was used (Kawaguchi et al., 1996) without describing its insight in the sense of Bahndorf (1991), Ströbel (1995) and von Scheven et al. (2021). The matrix presented in Jiang et al. (2012) for rigid-frame structures describes a different approach. Furthermore, inspired by the computation of the redundancy matrix in discrete representations of structures, von Scheven et al. (2021) also showed a novel procedure for the computation of redundancy distributions in continuous representations of one-dimensional structures. Its generalization to beam and multi-dimensional problems, however, does not appear to be straightforward. The latter work as well as (Bahndorf, 1991; Ströbel, 1995) is addressed in Section 2, as it forms the starting point of the present work.

In a continuum-mechanical setting of three-dimensional bodies, Kinoshita and Mura (1971), Furuhashi and Mura (1979) and Mura (1987) defined the notion "eigenstrains" as inelastic strains caused by e.g. thermal expansion, initial strains, plastic strains, or strains due to phase transformation. Following Hans Reissner's concept of *Eigenspannungen* ("Spannung" is German for "stress") published in Reissner (1931), stresses produced by these eigenstrains were referred to as "eigenstresses" (Mura, 1987). Assuming an inclusion of finite extent in a three-dimensional body of infinite extent, eigenstrain distributions that do not produce any stress (zero eigenstress) – so-called "impotent eigenstrains" – were derived Furuhashi and Mura (1979), based on Eshelby (1957). Irschik and Ziegler (2001) extended the concept of eigenstrains to structures with finite extent and with geometric boundary conditions and derived impotent eigenstrains in this setting. Moreover, they proposed the notion "self-stress" for stress distributions in structural problems produced by eigenstrains.

In the context of static shape control, the same authors worked out a direct method for computing the eigenstrain distribution that compensates (or at least minimizes) displacements from force loads (Irschik et al., 1997, 1999). The displacement at a certain structural point for a given eigenstrain distribution is computed by an integral formula for thermoelasticity (referred to as generalized Maysel's formula), using the appropriate influence function as integral kernel (Ziegler and Irschik, 1987; Irschik and Ziegler, 2001). Influence functions with respect to eigenstrains as well as Green's functions in (thermo)elastostatics, respectively, were investigated in e.g. Kinoshita and Mura (1971), Furuhashi and Mura (1979), Hartmann (1985), Ziegler and Irschik (1987), Irschik and Ziegler (2001), Grätsch (2002), Sheremet (2011), Hartmann (2013) and Xie et al. (2016). However, their relations to the concept of redundancy, to the best of our knowledge, are not scientifically investigated so far. This is part of the original contribution of the present work.

Furthermore, in Irschik and Ziegler (2001) the existence of eigenstrain distributions that induce stress without inducing deformation – so-called "nilpotent eigenstrains" – is shown. In the context of discrete and discretized representations (Ziegler, 2004), as well as of continuous representations (Nyashin et al., 2005), it is proven that the space of eigenstrain distributions may be uniquely decomposed into two orthogonal subspaces: Firstly, the subspace of stress-free impotent eigenstrains and secondly, the subspace of deformation-free nilpotent eigenstrains. The mentioned decomposition enables the independent adaptation of displacements and stresses in adaptive structures (Nyashin et al., 2005). These two subspaces are also addressed in Section 4.6 of the present work.

1.3. Outline

The paper is organized as follows: In Section 2 the concept of redundancy in discrete representations is reviewed (Section 2.1), based upon which the problem statement, the research hypothesis, and the limitations are elaborated (Section 2.2). Subsequently, in Section 3 the concept of Green's functions in linear elastostatics is given, also introducing the notation and definitions. In Section 4 the novel theoretical contribution of the present work is developed. Based on formulating the boundary value problem for the relevant influence functions in Section 4.1, the redundancy operator, the total self-strain operator, the self-stress operator, and the corresponding integral kernels are introduced, also providing their relations to Green's functions (Sections 4.2 to 4.3). Afterwards, the symmetry, trace, rank, and spectral properties are studied (Sections 4.4 to 4.7). Examples to confirm and to illustrate the theoretical findings are elaborated in Section 5. Finally, in Section 6 the article's results are concluded and possible directions of future work are given.

2. Redundancy distribution in discrete representations and problem statement

2.1. Review of concept of redundancy distribution

Consider a discrete, elastostatic representation of a structure, obeying linear truss theory, consisting of n_e truss elements ($i = 1, \dots n_e$)

and n_d degrees of freedom. $\mathbf{A}^\mathrm{T} \in \mathbb{R}^{n_\mathrm{d} \times n_\mathrm{e}}$ maps the vector of axial forces $\mathbf{N} \in \mathbb{R}^{n_\mathrm{e}}$ onto the vector of external nodal loads $\mathbf{f}_\mathrm{ext} \in \mathbb{R}^{n_\mathrm{d}}$ in the discrete equilibrium equation $\mathbf{A}^\mathrm{T}\mathbf{N} = \mathbf{f}_\mathrm{ext}$. $\mathbf{C} \in \mathbb{R}^{n_\mathrm{e} \times n_\mathrm{e}}$ maps the vector of elastic axial elongations $\Delta\mathbf{l}^\mathrm{el} \in \mathbb{R}^{n_\mathrm{e}}$ onto $\mathbf{N}$ in the constitutive equation $\mathbf{N} = \mathbf{C}\Delta\mathbf{l}^\mathrm{el}$. $\mathbf{A} = (\mathbf{A}^\mathrm{T})^\mathrm{T} \in \mathbb{R}^{n_\mathrm{e} \times n_\mathrm{d}}$ maps the vector of degrees of freedom $\mathbf{d} \in \mathbb{R}^{n_\mathrm{d}}$ onto the vector of total elemental elongations $\Delta\mathbf{l} \in \mathbb{R}^{n_\mathrm{e}}$ in the discrete kinematic equation $\Delta\mathbf{l} = \mathbf{A}\mathbf{d}$. $\Delta\mathbf{l}$ consists of an elastic part $\Delta\mathbf{l}^\mathrm{el}$ and an inelastic part $\Delta\mathbf{l}^0$, such that $\Delta\mathbf{l} = \Delta\mathbf{l}^\mathrm{el} + \Delta\mathbf{l}^0$. $\Delta\mathbf{l}^0$ can be interpreted mechanically as e.g. a temperature-induced initial elongation.

According to von Scheven et al. (2021), the redundancy relations read

$$-\Delta\mathbf{l}^\mathrm{el} = \underbrace{(-\mathbf{A}(\mathbf{A}^\mathrm{T}\mathbf{C}\mathbf{A})^{-1}\mathbf{A}^\mathrm{T}\mathbf{C} + \mathbf{I})}_{=:\,\mathbf{R}}\Delta\mathbf{l}^0, \quad \Delta\mathbf{l} = \underbrace{(-\mathbf{R} + \mathbf{I})}_{=:\,\mathbf{A}_0}\Delta\mathbf{l}^0. \tag{1}$$

Here, $\mathbf{R} \in \mathbb{R}^{n_\mathrm{e} \times n_\mathrm{e}}$ is the redundancy matrix, $\mathbf{A}_0 \in \mathbb{R}^{n_\mathrm{e} \times n_\mathrm{e}}$ is the total self-strain matrix and $\mathbf{I} \in \mathbb{R}^{n_\mathrm{e} \times n_\mathrm{e}}$ is the identity matrix. Considering Eq. (1), the redundancy matrix component R_{ik} maps the initial elongations imposed in element k onto the negative elastic elongations in element i. Therefore, in a truss structure with given support conditions and structural topology, R_{ik} describes the internal constraint acting on element i caused by element k. More generally speaking, $\mathbf{R}$ contains the spatial distribution of internal constraint within the structure, induced by redundancies with respect to the support conditions and/or the structural topology. This relation was revealed in Bahndorf (1991), considering the analogy between Gaussian adjustment calculus in geodesy and in structural mechanics. In geodetic networks a measurement of the distance between two points in space is redundant if it contributes to the reliability of the network coordinates to be determined. The more redundant a measurement is, the more it contributes, and vice versa. In structural mechanics of truss structures an element i is redundant, i.e. $R_{ii} > 0$, if it contributes to the statical indeterminacy, and thus, to the load carrying in a constraint load case imposed on this element ($\Delta l_i^0 \neq 0$). Equivalently, it undergoes internal constraint caused by the support conditions and/or the structural topology. The more redundant an element is, the more it contributes in the above-mentioned sense, and vice versa. An element i is non-redundant, i.e. $R_{ii} = 0$, if it does not contribute in the above-mentioned sense, and thus, it undergoes no internal constraint. However, its internal force is indispensible to satisfy equilibrium equations in an arbitrary force load case ($\mathbf{f}_\mathrm{ext} \neq \mathbf{0}$).

Reflecting the relations between redundancy, internal constraint and statical indeterminacy, it has also been shown that the distribution of internal constraint may equivalently be interpreted as the spatial distribution of the degree of statical indeterminacy n_s. Referring to Bahndorf (1991), Ströbel (1995) and von Scheven et al. (2021), the main-diagonal component R_{ii} is the contribution of element i to the degree of statical indeterminacy, such that

$$\mathrm{tr}_{\mathbb{R}^{n_\mathrm{e}}}(\mathbf{R}) = \sum_{i=1}^{n_\mathrm{e}} R_{ii} = n_\mathrm{s}, \tag{2}$$

with $\mathrm{tr}_{\mathbb{R}^{n_\mathrm{e}}}$ being the trace operator applied to quadratic matrices of dimension $n_\mathrm{e} \times n_\mathrm{e}$. The components R_{ii} are also known as distributed statical indeterminacy (see Chen et al., 2018; Zhou et al., 2015; Eriksson and Tibert, 2006).

Furthermore, it has been shown, that $\mathbf{R}$ and $\mathbf{A}_0$ have special properties. First, they are idempotent matrices, i.e. $\mathbf{R} = \mathbf{R}\mathbf{R}$, $\mathbf{A}_0 = \mathbf{A}_0\mathbf{A}_0$. Second, the relation $\mathbf{R} + \mathbf{A}_0 = \mathbf{I}$ (see Eq. (1)) holds. This implies that the eigenvalues of $\mathbf{R}$, denoted as λ_R, and the eigenvalues of $\mathbf{A}_0$, denoted as λ_{A_0}, are real, equal to either 1 or 0, and they are related by $\lambda_\mathrm{R} + \lambda_{\mathrm{A}_0} = 1$. The corresponding eigenvectors may be interpreted from a structural mechanics point of view. In von Scheven et al. (2021) further details in the context of discrete representations of truss and beam structures are given. In the context of continuous representations this is explored within the present work.

To bring the redundancy relations into a format applicable to any structural theory, it is reformulated in terms of strain quantities. Scaling Eq. (1) with the inverse length the relations yield

$$-\boldsymbol{\varepsilon}^\mathrm{el} = \mathbf{R}\boldsymbol{\varepsilon}^0, \quad \boldsymbol{\varepsilon} = \mathbf{A}_0\boldsymbol{\varepsilon}^0. \tag{3}$$

$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^\mathrm{el} + \boldsymbol{\varepsilon}^0 \in \mathbb{R}^{n_\mathrm{e}}$ is the vector of strain quantities, consisting of a part $\boldsymbol{\varepsilon}^\mathrm{el}$ caused by elastic deformation and an inelastic part $\boldsymbol{\varepsilon}^0$ caused by e.g. thermal expansion, pre-strain due to fabrication, or strains imposed by actuation. The latter part $\boldsymbol{\varepsilon}^0$ has the character of eigenstrains according to Furuhashi and Mura (1979) (see Section 1.2). However, as spectral properties (eigenvalues etc.) are investigated in later sections, $\boldsymbol{\varepsilon}^0$ is referred to as "initial strains" for the sake of clarity, remarking that its physical origin is not relevant within the present work. With respect to Eq. (3), R_{ik} maps the initial strain imposed in element k onto the negative elastic strain in element i. As equilibrium, constitutive, and kinematic equations in discrete representations of beam structures can also be expressed by using matrices $\mathbf{A}^\mathrm{T}$, $\mathbf{C}$, and $\mathbf{A}$ (see e.g. Przemieniecki, 1968; Krätzig et al., 2019; von Scheven et al., 2021), the redundancy relations can be formulated in the format of Eq. (3) as well.

2.2. Problem statement, research hypothesis, and limitations

To extend the findings from discrete to continuous representations, an operator notation is introduced. Considering Eq. (3) from an operator-theoretical perspective, the linear operator $\tilde{\mathcal{R}} : \mathbb{R}^{(n_\mathrm{e} n_\varepsilon)} \to \mathbb{R}^{(n_\mathrm{e} n_\varepsilon)}$ exists and is defined by

$$\tilde{\mathcal{R}}\boldsymbol{\varepsilon}^0 := \mathbf{R}\boldsymbol{\varepsilon}^0 = -\boldsymbol{\varepsilon}^\mathrm{el}. \tag{4}$$

It is introduced as redundancy operator for discrete, elastostatic, and linear representations of truss and beam and structures. Based on this perspective, the problem to solve in the present paper is twofold and stated as follows:

1. Find the operator that describes redundancy in continuous representations, i.e. find the operator $\mathcal{R} : C(\Omega, \mathbb{R}^{n_\varepsilon}) \to C(\Omega, \mathbb{R}^{n_\varepsilon})$ that maps the function of initial strain quantities onto the negative function of elastic strain quantities, i.e.

$$\mathcal{R}\boldsymbol{\varepsilon}^0(x) = -\boldsymbol{\varepsilon}^\mathrm{el}(x); \tag{5}$$

2. study properties of this operator to constitute a continuum-mechanical theory of redundancy in structures.

$C(\Omega, \mathbb{R}^{n_\varepsilon})$ is the function space of all sufficiently continuously-differentiable, vector-valued functions defined on the domain Ω and in the vector space $\mathbb{R}^{n_\varepsilon}$, respectively. The notion "continuum-mechanical" refers to the representations being spatially continuous.

In Eq. (4), the part $(\mathbf{A}^\mathrm{T}\mathbf{C}\mathbf{A})^{-1}\mathbf{A}^\mathrm{T}\mathbf{C}$ is identified as the inverse stiffness matrix (also known as flexibility matrix) multiplied by a matrix of column-wise arranged load vectors. The continuum-mechanical counterpart of the flexibility matrix is the Green's function of the corresponding boundary-value problem. Green's function is a special influence function, namely the influence function of a displacement quantity with respect to an external load quantity. This gives rise to the following research hypothesis:

> The redundancy map in continuous representations is provided by the negative influence functions of the elastic strain quantities with respect to initial strain quantities.

The present work is applicable to continuous representations of structures possessing the following two properties: First, the underlying structural theory is statically determinate. Second, it obeys static equilibrium, linear kinematics as well as linear elastic constitutive law. Appropriate exemplary structural theories are: 1d rod theory, 2d-Euler-Bernoulli and 2d-Bresse-Timoshenko beam theories, circularly curvilinear plane Euler-Bernoulli beam theory under out-of-plane loading, as well as membrane theory of axisymmetric Kirchhoff-Love shells under axisymmetric loading. Both constant and variable cross sections or shell thickness, respectively, may be considered.

3. Structural theories and Green's functions

3.1. Boundary-value problem of linear elastostatics

For an infinitesimal element in the one-dimensional open domain $\Omega = (\Gamma_{\mathrm{lb}}, \Gamma_{\mathrm{ub}})$ to obey an arbitrary linear elastostatic structural theory, the system of differential and algebraic equations in the variable $x \in \Omega$ reads

$$\mathcal{L}^{\mathrm{x}*}_{mj}\sigma_j(x) = p_m(x), \qquad m = 1,\dots,M, \quad \text{(equilibrium)} \tag{6a}$$

$$\sigma_j(x) = C_{ji}(x)(\varepsilon_i(x) - \varepsilon^0_i(x)), \quad j = 1,\dots,J, \quad \text{(constitutive law)} \tag{6b}$$

$$\varepsilon_i(x) = \mathcal{L}^{\mathrm{x}}_{in}u_n(x), \qquad i = 1,\dots,J. \quad \text{(kinematic equation)} \tag{6c}$$

Here and throughout the present work Einstein summation convention is applied to compositions of two quantities with italic, latin subscripted indices. u_n, σ_j, and p_m are components of vector-valued functions of displacement quantities $\mathbf{u} \in \mathbb{R}^M$, stress quantities $\boldsymbol{\sigma} \in \mathbb{R}^J$, and distributed external load quantities $\mathbf{p} \in \mathbb{R}^M$, respectively. C_{ji} is a component of the constitutive matrix $\mathbf{C} \in \mathbb{R}^{J\times J}$. ε_i is a component of the vector-valued function of strain quantities $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^{\mathrm{el}} + \boldsymbol{\varepsilon}^0 \in \mathbb{R}^J$. It consists of elastic strain quantities (superscript el) and initial strain quantities (superscript 0), where the latter may be caused by e.g. thermal expansion, prestrain due to fabrication, or actuation. $\mathcal{L}^{\mathrm{x}}_{in} : C(\Omega,\mathbb{R}) \to C(\Omega,\mathbb{R})$ and $\mathcal{L}^{\mathrm{x}*}_{mj} : C(\Omega,\mathbb{R}) \to C(\Omega,\mathbb{R})$ are components of the linear, matrix-valued differential operator $\mathcal{L}^{\mathrm{x}} : C(\Omega,\mathbb{R}^M) \to C(\Omega,\mathbb{R}^J)$ and its adjoint operator $\mathcal{L}^{\mathrm{x}*} : C(\Omega,\mathbb{R}^J) \to C(\Omega,\mathbb{R}^M)$, respectively (acting on x).

Eqs. (6) form a system of $M+2J$ differential and algebraic equations in the unknown functions $\boldsymbol{\sigma}$, $\boldsymbol{\varepsilon}$ and $\mathbf{u}$. Here, $M = J$ (i.e. $\mathcal{L}^{\mathrm{x}*}$ is square), as statically determinate structural theories are considered. Elimination of $\boldsymbol{\sigma}$ and $\boldsymbol{\varepsilon}$ from Eqs. (6) yields a set of M differential equations in $\mathbf{u}$ only:

$$\underbrace{\mathcal{L}^{\mathrm{x}*}_{mj}C_{ji}(x)\mathcal{L}^{\mathrm{x}}_{in}}_{=:\,D^{\mathrm{x}}_{mn}}u_n(x) = p_m(x) + \underbrace{\mathcal{L}^{\mathrm{x}*}_{mj}C_{ji}(x)}_{=:\,\mathcal{P}^{\mathrm{x}}_{mi}}\varepsilon^0_i(x), \quad x \in \Omega, \quad m = 1,\dots,M. \tag{7a}$$

$D^{\mathrm{x}}_{mn} : C(\Omega,\mathbb{R}) \to C(\Omega,\mathbb{R})$ and $\mathcal{P}^{\mathrm{x}}_{mi} : C(\Omega,\mathbb{R}) \to C(\Omega,\mathbb{R})$ are components of linear operators acting on x. D^{x}_{mn} is self-adjoint (Arfken et al., 2013, p. 1070). The Q_{u} Dirichlet boundary conditions on Γ_{u} and the Q_σ Neumann boundary conditions on Γ_σ read

$$\mathcal{Q}^{\mathrm{x}}_{qn}u_n(x)\Big|_{x=x_q} = 0, \quad x_q \in \Gamma_\sigma \cup \Gamma_{\mathrm{u}}, \quad q = 1,\dots,Q, \quad Q := Q_{\mathrm{u}} + Q_\sigma. \tag{7b}$$

$\mathcal{Q}_{qn} : C(\Omega,\mathbb{R}) \to C(\Omega,\mathbb{R})$ is a component of a linear operator acting on x forming the left-hand sides of the homogeneous boundary conditions. Considering inhomogeneous boundary conditions would lead to boundary terms in subsequent integral expressions. However, the operators and kernels studied in Section 4, as well as its properties, would not change compared to the case with homogeneous boundary conditions.

Eqs. (7a) and (7b) form a boundary-value problem (BVP) of a structure loaded by distributed external load quantities $\mathbf{p}$ and distributed initial strain quantities $\boldsymbol{\varepsilon}^0$. $\mathbf{p}$ as well as $\boldsymbol{\varepsilon}^0$ are assumed to be sufficiently continuously-differentiable such that all occurring differentials exist. The relevant quantities and equations for particular statically determinate structural theories are arranged in Table 1.

3.2. Green's functions in linear elastostatics

There exist M sets of M scalar-valued functions $G_{nb} : \Omega^2 \to \mathbb{R}$, $n = 1,\dots,M$, each set of which solves the following set of M differential equations and satisfies the Q corresponding homogeneous boundary conditions:

$$D^{\mathrm{x}}_{mn}G_{nb}(x,s) = \tilde{\delta}_{mb}\delta(s-x), \quad x,s \in \Omega, \qquad m = 1,\dots,M, \quad b = 1,\dots,M, \tag{8a}$$

$$\mathcal{Q}^{\mathrm{x}}_{qn}G_{nb}(x,s)\Big|_{x=x_q} = 0, \qquad x_q \in \Gamma_\sigma \cup \Gamma_{\mathrm{u}}, \quad q = 1,\dots,Q. \tag{8b}$$

G_{nb} is the scalar-valued Green's function and a component of the matrix-valued function $\mathbf{G} : \Omega^2 \to \mathbb{R}^{M\times M}$. $\tilde{\delta}_{mb}$ is the Kronecker delta. $\delta(s-x)$ is the Dirac delta function, which is a function in a distribution-theoretical sense. Relevant properties are its symmetry, the filtering property, and its idempotence (Arfken et al., 2013, p. 75ff.). From a mathematical point of view, the set of Green's functions solves the BVP (8). Once it is known, the solution of the BVP for distributed external load quantities ($\mathbf{p} \neq \mathbf{0}$, $\boldsymbol{\varepsilon}^0 = \mathbf{0}$) can be represented as an integral:

$$u_n(x) = \int_\Omega G_{nb}(x,s)p_b(s)\,\mathrm{d}s. \tag{9}$$

From a mechanical point of view, $G_{nb}(x,s)$ is the displacement quantity u_n at x (observation point) caused by a Dirac impulse in the distributed external load quantity p_b imposed at s (source point), which is equal to a single load imposed at s of magnitude $\int_\Omega \delta(s-x)\,\mathrm{d}s = 1$. In this light, Green's function is identified as the flexibility function. Thus, the graph of Green's function is identical to the influence line for displacement u_n known from classical structural mechanics.

Symmetry properties of Green's functions

Green's functions in elastostatics are, in general, symmetric in the following sense:

$$G_{nb}(x,s) = G_{nb}(s,x), \quad \forall\, n,b \quad \Leftrightarrow \quad \mathbf{G}(x,s) = \mathbf{G}(s,x), \tag{10a}$$

$$G_{nb}(x,s) = G_{bn}(s,x), \quad \forall\, n,b \quad \Leftrightarrow \quad \mathbf{G}(x,s) = \mathbf{G}^{\mathrm{T}}(s,x), \tag{10b}$$

$$G_{nb}(x,s) = G_{bn}(s,x), \quad \forall\, n,b \quad \Leftrightarrow \quad \mathbf{G}(x,s) = \mathbf{G}^{\mathrm{T}}(x,s). \tag{10c}$$

Eq. (10a) means invariance with respect to transposition of observation and source point, which follows directly from the symmetry of the Dirac delta function. Eq. (10b) means invariance with respect to transposition of observation and source point and with respect to transposition of components of displacement and load quantities. This is reflected by Betti's reciprocity theorem. Eq. (10c) means invariance with respect to transposition of components of displacement and load quantities, which follows from Eqs. (10a) and (10b) by transitivity. Eqs. (10) are proven in Appendix A.1.

4. Continuum-mechanical theory of redundancy

4.1. Influence functions for displacement quantities

There exist J sets of M scalar-valued influence functions $E_{\mathrm{u}_n,\varepsilon^0_k} : \Omega^2 \to \mathbb{R}$, $n = 1,\dots,M$, each set of which solves the following set of M differential equations and satisfies the Q corresponding homogeneous boundary conditions:

$$D^{\mathrm{x}}_{mn}E_{\mathrm{u}_n,\varepsilon^0_k}(x,s) = \mathcal{P}^{\mathrm{x}}_{mi}\tilde{\delta}_{ik}\delta(s-x), \quad x,s \in \Omega, \qquad m = 1,\dots,M, \quad k = 1,\dots,J, \tag{11a}$$

$$\mathcal{Q}^{\mathrm{x}}_{qn}E_{\mathrm{u}_n,\varepsilon^0_k}(x,s)\Big|_{x=x_q} = 0, \qquad x_q \in \Gamma_\sigma \cup \Gamma_{\mathrm{u}}, \quad q = 1,\dots,Q. \tag{11b}$$

From a mathematical point of view, the set of influence functions solves the BVP (11). Once it is known, the solution of the BVP (7) for distributed initial strain quantities ($\mathbf{p} = \mathbf{0}$, $\boldsymbol{\varepsilon}^0 \neq \mathbf{0}$) can be represented as an integral:

$$u_n(x) = \int_\Omega E_{\mathrm{u}_n,\varepsilon^0_k}(x,s)\varepsilon^0_k(s)\,\mathrm{d}s, \quad E_{\mathrm{u}_n,\varepsilon^0_k}(x,s) = C_{pk}(s)\mathcal{L}^{\mathrm{s}}_{pb}G_{nb}(x,s). \tag{12}$$

From a mechanical point of view, $E_{\mathrm{u}_n,\varepsilon^0_k}(x,s)$ is the displacement quantity u_n at x caused by a Dirac impulse in the initial strain quantity ε^0_k imposed at s. A mechanical interpretation of a Dirac impulse on initial strain quantities in different structural theories is exemplarily given in Table 2.

Table 1
Overview of quantities and operators for particular statically determinate structural theories.

Structural theory	Displacement quantities in $\mathbf{u}(x)$	Strain quantities in $\boldsymbol{\varepsilon}(x)$	Stress quantities in $\boldsymbol{\sigma}(x)$	External load quantities in $\mathbf{p}(x)$	Adjoint differential operator $\mathcal{L}^{x*}$	Constitutive matrix $\mathbf{C}(x)$	Differential operator $\mathcal{L}^{x}$
1d rod	Axial displacement u	Axial strain ε	Normal force N	Axial force n	$-\frac{\mathrm{d}}{\mathrm{d}x}$	EA	$\frac{\mathrm{d}}{\mathrm{d}x}$
2d-Euler-Bernoulli beam	Transverse displacement w	Curvature κ	Bending moment M	$q+\frac{\mathrm{d}m}{\mathrm{d}x}$ Transverse force q bending moment m	$-\frac{\mathrm{d}^2}{\mathrm{d}x^2}$	EI	$-\frac{\mathrm{d}^2}{\mathrm{d}x^2}$
2d-Bresse-Timoshenko beam[a]	Cross-sectional rotation φ transverse displacement w	Curvature κ shear angle γ	Bending moment M shear force V	Bending moment m transverse force q	$\begin{bmatrix} -\frac{\mathrm{d}}{\mathrm{d}x} & 1 \\ 0 & -\frac{\mathrm{d}}{\mathrm{d}x} \end{bmatrix}$	$\begin{bmatrix} EI & 0 \\ 0 & GA_\mathrm{S} \end{bmatrix}$	$\begin{bmatrix} \frac{\mathrm{d}}{\mathrm{d}x} & 0 \\ 1 & \frac{\mathrm{d}}{\mathrm{d}x} \end{bmatrix}$
Circularly curvilinear plane Euler-Bernoulli beam in arc-length coordinates (out-of-plane loading)	Out-of-plane displacement w torque rotation θ	Curvature κ drilling ϑ	Bending moment M torque moment M_T	Out-of-plane force q torque moment m_T	$\begin{bmatrix} -\frac{\mathrm{d}^2}{\mathrm{d}x^2} & \frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{l} \\ -\frac{1}{l} & -\frac{\mathrm{d}}{\mathrm{d}x} \end{bmatrix}$	$\begin{bmatrix} EI & 0 \\ 0 & GI_\mathrm{T} \end{bmatrix}$	$\begin{bmatrix} -\frac{\mathrm{d}^2}{\mathrm{d}x^2} & -\frac{1}{l} \\ -\frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{l} & \frac{\mathrm{d}}{\mathrm{d}x} \end{bmatrix}$
	EA: axial stiffness EI: bending stiffness GA_S: shear stiffness GI_T: torque stiffness	x: length or arc-length coordinate l = const.: radius of beam axis					

[a]According to Challamel and Elishakoff (2019) the shear-deformable beam theory usually associated with Timoshenko (1916) only, is also associated to Bresse, who derived the equations before (Bresse, 1859), however without considering the concept of shear correction.

Table 2
Mechanical interpretation of Dirac impulse on initial strain quantities.

Structural theory	Initial strain quantities	Jump condition	Mechanical interpretation
1d rod	$\varepsilon^0(x) := \delta(s-x)$	$[[u(s)]] = \int_\Omega \delta(s-x)\,\mathrm{d}s = 1$	Jump in u of 1 at s
2d Euler-Bernoulli beam	$\kappa^0(x) := \delta(s-x)$	$[[\varphi(s)]] = \int_\Omega \delta(s-x)\,\mathrm{d}s = 1$	Jump in φ of 1 at s
2d Bresse-Timoshenko beam	$\gamma^0(x) := 0$ $\kappa^0(x) := \delta(s-x)$	$[[w_\mathrm{S}(s)]] = 0$ $[[\varphi(s)]] = \int_\Omega \delta(s-x)\,\mathrm{d}s = 1$	Continuity in w_S Jump in φ of 1 at s
	$\kappa^0(x) := 0$ $\gamma^0(x) := \delta(s-x)$	$[[\varphi(s)]] = 0$ $[[w_\mathrm{S}(s)]] = \int_\Omega \delta(s-x)\,\mathrm{d}s = 1$	Continuity in φ Jump in w_S of 1 at s

Legend: $[[(\cdot)(s)]] := \lim_{\epsilon\to 0}((\cdot)(s+e)-(\cdot)(s-e))$
$w_\mathrm{S}(x) := \int_{\Gamma_\mathrm{lb}}^{x} \gamma(\bar{x})\mathrm{d}\bar{x}$: shear displacement

4.2. Redundancy operator and total self-strain operator

Applying $\mathcal{L}^{\mathrm{x}}_{in}$ to Eq. (12) and using kinematic Eqs. (6c), the total strains caused by distributed initials strains as an integral is obtained:

$$\varepsilon_i(x) = \mathcal{L}^{\mathrm{x}}_{in} u_n(x) = \int_\Omega \underbrace{\left(\mathcal{L}^{\mathrm{x}}_{in} C_{pk}(s) \mathcal{L}^{\mathrm{s}}_{pb} G_{nb}(x,s)\right)}_{=:\, E_{\varepsilon_i,\varepsilon^0_k}(x,s) =:\, A_{0ik}(x,s)} \varepsilon^0_k(s)\,\mathrm{d}s. \tag{13}$$

$\mathbf{A}_0 : \Omega^2 \to \mathbb{R}^{J\times J}$ is introduced as total self-strain function. It is the matrix-valued integral kernel of the integral operator $\mathcal{A}_0 : C(\Omega,\mathbb{R}^J) \to C(\Omega,\mathbb{R}^J)$ defined by

$$\mathcal{A}_0\boldsymbol{\varepsilon}^0(x) := \int_\Omega \mathbf{A}_0(x,s)\boldsymbol{\varepsilon}^0(s)\,\mathrm{d}s = \boldsymbol{\varepsilon}(x). \tag{14}$$

Here and throughout the present work an integral of a matrix-valued function is applied component-wise. $\mathcal{A}_0$ is introduced as total self-strain operator. The name is chosen, because it maps initial strain distributions onto total strain distributions, the corresponding stress states of which are equilibrated within the structure itself, i.e. in the absence of external load quantities. The total self-strain operator is the continuum-mechanical counterpart of the matrix $\mathbf{A}_0$ introduced in Bahndorf (1991) and further studied in Ströbel (1995) and von Scheven et al. (2021).

Using Eq. (13) and considering $\varepsilon_i = \varepsilon^0_i + \varepsilon^{\mathrm{el}}_i$, yields the negative elastic strains caused by distributed initial strains as an integral:

$$-\varepsilon^{\mathrm{el}}_i(x) = \int_\Omega \underbrace{\left(-\mathcal{L}^{\mathrm{x}}_{in} C_{pk}(s) \mathcal{L}^{\mathrm{s}}_{pb} G_{nb}(x,s) + \bar{\delta}_{ik}\delta(s-x)\right)}_{=-A_{0ik}(x,s)+\bar{\delta}_{ik}\delta(s-x) \,=:\, -E_{\varepsilon^{\mathrm{el}}_i,\varepsilon^0_k}(x,s) \,=:\, R_{ik}(x,s)} \varepsilon^0_k(s)\,\mathrm{d}s. \tag{15}$$

Finally, the influence function $\mathbf{R} : \Omega^2 \to \mathbb{R}^{J\times J}$ is introduced as negative elastic self-strain function or redundancy function. It is the matrix-valued kernel of the integral operator $\mathcal{R} : C(\Omega,\mathbb{R}^J) \to C(\Omega,\mathbb{R}^J)$:

$$\mathcal{R}\boldsymbol{\varepsilon}^0(x) := \int_\Omega \mathbf{R}(x,s)\boldsymbol{\varepsilon}^0(s)\,\mathrm{d}s = -\boldsymbol{\varepsilon}^{\mathrm{el}}(x). \tag{16}$$

$\mathcal{R}$ is introduced as negative elastic self-strain operator or redundancy operator. The first name is chosen, because the operator maps initial strain distributions onto negative elastic strain distributions, the corresponding stress states of which are equilibrated within the structure itself, i.e. in the absence of external load quantities. This operator is the continuum-mechanical counterpart of the redundancy matrix described in Bahndorf (1991), Ströbel (1995) and von Scheven et al. (2021) (compare Eqs. (4) and (15)), which motivates the second name "redundancy operator". Therefore, the research hypothesis in Section 2.2 is verified and the generalization of the redundancy concept to the continuous level is achieved.

Considering Eq. (15), the relation between total self-strain and redundancy operator yields

$$\mathcal{R} + \mathcal{A}_0 = \mathcal{I}, \tag{17}$$

where $\mathcal{I} : C(\Omega,\mathbb{R}^J) \to C(\Omega,\mathbb{R}^J)$ is the identity operator. This relation is analogous to the discrete version, Eq. (1). The redundancy operator and the total self-strain operator as well as the corresponding maps are idempotent (proof in Appendix A.2).

4.3. Self-stress operator

Applying constitutive law (Eq. (6b)) to Eq. (15) yields the stress quantities caused by distributed initial strains as an integral:

$$\sigma_j(x) = \int_\Omega \underbrace{-C_{ji}(x)\left(-\mathcal{L}^{\mathrm{x}}_{in}C_{pk}(s)\mathcal{L}^{\mathrm{s}}_{pb}G_{nb}(x,s) + \tilde{\delta}_{ik}\delta(s-x)\right)}_{=-C_{ji}(x)R_{ik}(x,s)\,=:\,E_{\sigma_j,\varepsilon^0_k}(x,s)\,=:\,S_{jk}(x,s)} \varepsilon^0_k(s)\,\mathrm{d}s. \tag{18}$$

The influence function $\mathbf{S} : \Omega^2 \to \mathbb{R}^{J\times J}$ is introduced as self-stress function. It is the matrix-valued integral kernel of the integral operator $S : C(\Omega,\mathbb{R}^J) \to C(\Omega,\mathbb{R}^J)$ defined by

$$S\boldsymbol{\varepsilon}^0(x) := \int_\Omega \mathbf{S}(x,s)\boldsymbol{\varepsilon}^0(s)\,\mathrm{d}s = \boldsymbol{\sigma}(x). \tag{19}$$

S is introduced as self-stress operator for spatially continuous, elastostatic and linear representations of structures. It maps initial strain distributions onto stress distributions that are equilibrated within the structure itself, i.e. in the absence of external load quantities.

4.4. Symmetry properties of self-stress function

The self-stress function is, in general, symmetric in the following sense:

$$S_{ik}(x,s) = S_{ki}(s,x), \quad \forall\, i,k \quad \Leftrightarrow \quad \mathbf{S}(x,s) = \mathbf{S}^{\mathrm{T}}(s,x). \tag{20}$$

From a mathematical point of view, this symmetry is obvious by considering Eq. (18). From a mechanical point of view, it is reflected by the fact that the source (Dirac impulse on distributed initial strain component) and the response (stress component) are energetically conjugate. A proof of this symmetry based on Betti's reciprocity theorem is given in Appendix A.3.

4.5. Redundancy densities, redundancy contributions and trace properties

The redundancy function component R_{ik} for $s = x$, which equals the influence function $-E_{\varepsilon^{\mathrm{el}}_i,\varepsilon^0_k}$ for observation and source point being identical, reads

$$R_{ik}(x,x) = -E_{\varepsilon^{\mathrm{el}}_i,\varepsilon^0_k}(x,x) =: \bar{R}_{ik}(x). \tag{21}$$

$\bar{R}_{ik}(x)$ for $k = i$ is denoted as redundancy distribution or redundancy density for the strain type i. It gives the spatial distribution of the redundancy in the structure contributed by this strain type. $\bar{R}_{ik}(x)$ for $k \neq i$ gives the spatial distribution of the coupling of the strain types i and k within the structure (e.g. for a Bresse-Timoshenko beam structure $i = \{\kappa,\gamma\}$, $k = \{\kappa,\gamma\}$, Table 1). In general $\bar{R}_{ik}(x) \neq \bar{R}_{ki}(x)$, for $i \neq k$ holds. Thus, the redundancy density, originally defined in von Scheven et al. (2021) for continuous representations of 1d structures, is generalized.

As $\mathcal{R}$ and $\mathcal{R}_{ik}$ are integral operators, they are trace-class operators (Trèves, 1967). Thus, the trace of the redundancy operator component $\mathcal{R}_{ik}$ is given as follows (Brislawn, 1988; Grieser, 2007):

$$\mathrm{tr}_\Omega(\mathcal{R}_{ik}) = \int_\Omega \bar{R}_{ik}(x)\,\mathrm{d}x =: r_{ik}. \tag{22}$$

r_{ik} is an aggregated measure for the coupling of the strain types i and k, for $i \neq k$. The trace of the redundancy operator $\mathcal{R}$ is

$$\mathrm{tr}_\Omega(\mathcal{R}) := \sum_i \int_\Omega \bar{R}_{ii}(x)\,\mathrm{d}x = \sum_i r_{ii} = n_{\mathrm{s}} =: r. \tag{23}$$

r_{ii} is the contribution of the strain type i to the total redundancy r of the structure. $\mathrm{tr}_\Omega(\mathcal{R})$ is equal to the degree of statical indeterminacy of the structure n_{s}, which is shown in the following subsection. As statically determinate structural theories are considered, r is a finite, non-negative integer number.

4.6. Spectral properties of redundancy operator and total self-strain operator

The spectral properties of the redundancy operator are studied from the eigenvalue equation

$$\mathcal{R}\boldsymbol{\phi}_h(x) = \int_\Omega \mathbf{R}(x,s)\boldsymbol{\phi}_h(s)\,\mathrm{d}s = \lambda_h\boldsymbol{\phi}_h(x), \quad h = 1,\dots,\infty, \tag{24}$$

where $\boldsymbol{\phi}_h : \Omega \to \mathbb{R}^J$ is the hth vector of eigenfunctions and λ_h the corresponding eigenvalue of $\mathcal{R}$. Assuming the redundancy function to be a separable kernel, i.e.

$$\mathbf{R}(x,s) = \sum_{a=1}^{A} \mathbf{v}_a(x)\mathbf{q}_a{}^{\mathrm{T}}(s), \quad \mathbf{v}_a,\mathbf{q}_a : \Omega \to \mathbb{R}^J, \tag{25}$$

the eigenvalue equation can be traced back to matrix eigenvalue problem of finite size (Appendix A.4). The separation in Eq. (25) as well as the number of summands A are, in general, non-unique. However, there exists a unique integer number $L \leq A$ being the least number of summands to represent $\mathbf{R}$, i.e.

$$\mathbf{R}(x,s) = \sum_{l=1}^{L} \mathbf{r}_l(x)\mathbf{t}_l{}^{\mathrm{T}}(s), \quad \mathbf{r}_l,\mathbf{t}_l : \Omega \to \mathbb{R}^J. \tag{26}$$

Eq. (26) is referred to as "minimal representation" (Gohberg and Kaashoek, 1987) of the integral kernel $\mathbf{R}$. As $\mathbf{R}$ contains negative unit elastic strain distributions (compare Eq. (16)), the number of summands to form the minimal representation is equal to the degree of statical indeterminacy, i.e.

$$L = n_{\mathrm{s}}. \tag{27}$$

Furthermore, L gives the dimension of the image of the corresponding integral operator $\mathcal{R}$, which is equal to its rank (Heil, 2018, p. 244):

$$\mathrm{rank}(\mathcal{R}) = \dim(\mathrm{im}(\mathcal{R})) = L. \tag{28}$$

As $\mathcal{R}$ is an integral operator, which fulfills the conditions of being a trace-class operator in a separable Hilbert space, i.e. $\mathrm{tr}_\Omega(\mathcal{R})$ is finite, Lidskii's theorem yields the following relation between the trace and the eigenvalues of $\mathcal{R}$ (Einsiedler and Ward, 2017, p. 193; Trèves, 1967):

$$\mathrm{tr}_\Omega(\mathcal{R}) = \sum_{n=1}^{\infty} \lambda_n. \tag{29}$$

As the redundancy operator $\mathcal{R}$ is idempotent (Appendix A.2), L eigenvalues equal 1, and an infinite number of eigenvalues equal 0. Thus, Eq. (29) can be reduced to a sum from 1 to L and is thus equal to L:

$$\mathrm{tr}_\Omega(\mathcal{R}) = \sum_{l=1}^{L} \lambda_l = L. \tag{30}$$

The set of Eqs. (23), (27), (28), and (30) is a crucial result for constituting redundancy theory as it associates spectral, trace, space, and rank properties of the redundancy operator with the degree of statical indeterminacy. Therefore, the properties of the redundancy matrix studied in Bahndorf (1991), Ströbel (1995) and von Scheven et al. (2021) are extended to the spatially continuous level.

In the sequel, the mechanical interpretation of the eigenfunctions is developed. Taking Eq. (17) into account and using filtering property of the Dirac delta function, the eigenvalue equations for the redundancy operator and for the total self-strain operator are related as follows:

$$\lambda_h\boldsymbol{\phi}_h(x) = \mathcal{R}\boldsymbol{\phi}_h(x), \tag{31a}$$

$$(1-\lambda_h)\boldsymbol{\phi}_h(x) = \mathcal{A}_0\boldsymbol{\phi}_h(x), \quad h = 1,\dots,\infty. \tag{31b}$$

Thus, redundancy operator and total self-strain operator share the same eigenfunctions. In conclusion, the image spaces and null spaces of the redundancy operator and of the total self-strain operator are

$$\mathrm{im}(\mathcal{R}) = \mathrm{null}(\mathcal{A}_0) = \mathrm{span}\{\boldsymbol{\phi}_h(x).\boldsymbol{\phi}_h(s) \mid \mathcal{R}\boldsymbol{\phi}_h = \lambda_h\boldsymbol{\phi}_h,\ \lambda_h = 1\}, \tag{32a}$$

$$\mathrm{null}(\mathcal{R}) = \mathrm{im}(\mathcal{A}_0) = \mathrm{span}\{\boldsymbol{\phi}_h(x).\boldsymbol{\phi}_h(s) \mid \mathcal{R}\boldsymbol{\phi}_h = \lambda_h\boldsymbol{\phi}_h,\ \lambda_h = 0\}. \tag{32b}$$

Considering Eqs. (16), (14), and (31), $\boldsymbol{\phi}_h$ may be interpreted as initial strain distribution $\boldsymbol{\varepsilon}^0$, $\lambda_h\boldsymbol{\phi}_h$ as negative elastic strain distribution $-\boldsymbol{\varepsilon}^{\mathrm{el}}$, and $(1-\lambda_h)\boldsymbol{\phi}_h$ as total strain distribution $\boldsymbol{\varepsilon}$. Adding Eqs. (31a) and (31b) yields

$$\boldsymbol{\phi}_h(x) = \mathcal{R}\boldsymbol{\phi}_h(x) + \mathcal{A}_0\boldsymbol{\phi}_h(x) \tag{33a}$$

$$= \lambda_h\boldsymbol{\phi}_h(x) + (1-\lambda_h)\boldsymbol{\phi}_h(x), \quad h = 1, \dots, \infty. \tag{33b}$$

As λ_h is either 1 or 0, respectively, Eq. (33) may be interpreted as decomposition of initial strain distribution into either purely non-compatible ($\boldsymbol{\varepsilon}^{\mathrm{el}} \neq \mathbf{0}$) and deformation-free ($\boldsymbol{\varepsilon} = \mathbf{0} \Rightarrow \mathbf{u} = \text{const.}$) initial strain distribution ($\in \mathrm{im}(\mathcal{R}) = \mathrm{null}(\mathcal{A}_0)$) or purely compatible ($\boldsymbol{\varepsilon}^{\mathrm{el}} = \mathbf{0}$) and deformation-inducing ($\boldsymbol{\varepsilon} \neq \mathbf{0}$) initial strain distribution ($\in \mathrm{null}(\mathcal{R}) = \mathrm{im}(\mathcal{A}_0)$) , respectively.

Finally, conclusions made for matrices $\mathbf{R}$ and $\mathbf{A}_0$ in von Scheven et al. (2021) may be extended to the spatially continuous level (also refer to Table 3): $\mathrm{im}(\mathcal{R})$ spans an n_{s}-dimensional space of non-compatible initial strain distributions in a structure. Initial strain distributions from this subspace will lead to zero deformations of the structure as $\boldsymbol{\varepsilon} = \mathbf{0} \Leftrightarrow \mathbf{u} = \text{const.}$, but to stress distribution that are equilibrated within the structure itself (i.e. in the absence of external load quantities) as $\boldsymbol{\varepsilon}^{\mathrm{el}} \neq \mathbf{0}$ (non-compatible deformations). This subspace is the complete space of the so-called "nilpotent eigenstrains" described by e.g. Irschik and Ziegler (2001) and Nyashin et al. (2005). On the contrary, $\mathrm{im}(\mathcal{A}_0)$ spans the infinite-dimensional space of compatible initial strain distributions. I.e. initial strain distributions from this subspace lead only to non-zero total strain distribution $\boldsymbol{\varepsilon} \neq \mathbf{0}$, but yield zero stresses as $\boldsymbol{\varepsilon}^{\mathrm{el}} = \mathbf{0}$. This subspace is the complete space of the so-called "impotent eigenstrains" described by e.g. Furuhashi and Mura (1979), Irschik and Ziegler (2001) and Nyashin et al. (2005).

4.7. Spectral properties of self-stress operator

If the redundancy function is a separable kernel and the redundancy operator is of finite rank, i.e. $\mathrm{rank}(\mathcal{R}) = L = n_{\mathrm{s}}$, then so is the self-stress function and operator, respectively. If so, the eigenfunctions of $\mathcal{S}$, $\boldsymbol{\psi}_h : \Omega \to \mathbb{R}^J$ and eigenvalues η_h, may be computed in the same manner as described in Section 4.6 and Appendix A.4. They solve the eigenvalue equation

$$\mathcal{S}\boldsymbol{\psi}_h(x) = \int_\Omega \mathbf{S}(x,s)\boldsymbol{\psi}_h(s)\,\mathrm{d}s = \eta_h\boldsymbol{\psi}_h(x), \quad h = 1, \dots, \infty. \tag{34}$$

It is assumed that the eigenfunctions are scaled such that they fulfill the L^2 normalization condition $\|\boldsymbol{\psi}_h(x)\|_2 = 1$. As the self-stress function is symmetric (Eq. (20)), it can be represented by

$$\mathbf{S}(x,s) = \sum_{h=1}^{\infty} \eta_h\boldsymbol{\psi}_h(x)\boldsymbol{\psi}_h(s) = \sum_{l=1}^{L} \eta_l\boldsymbol{\psi}_l(x)\boldsymbol{\psi}_l(s), \tag{35}$$

with $\eta_{(\cdot)} \in \mathbb{R}$ and with the normalized eigenfunctions forming an orthonormal basis. This states the self-stress operator $\mathcal{S}$ being self-adjoint (Arfken et al., 2013, p. 1070).

Considering Eqs. (19), (34), and (35), the normalized eigenfunctions may be interpreted as orthonormal distributions of initial strain quantities, whereas the eigenvalues may be considered as corresponding modal self-stress quantities. Thus, distributions from the image space $\mathrm{im}(\mathcal{S})$ (is spanned by $\eta_l\boldsymbol{\psi}_l$, for $l = 1, \dots, L$) may be seen as orthogonal distributions of stress quantities equilibrated in the structure itself, i.e. in the absence of external load quantities. Such stress distributions are referred to as self-stress distributions in the sense of Irschik/Ziegler (e.g. Irschik and Ziegler, 2001). $\mathrm{im}(\mathcal{S})$ form the continuum-mechanical counterpart of the vector space of self-stress states described in e.g. Pellegrino and Calladine (1986), Pellegrino (1990) and Pellegrino (1993).

5. Examples

Subsequently, five exemplary continuous representations of structures are investigated. More precisely, examples of a fixed 1d rod (Section 5.1), a clamped 2d-Bresse-Timoshenko beam (Section 5.2), two different plane frames (Sections 5.3 and 5.4), and a plane quarter-circular Euler-Bernoulli beam (Section 5.5) are worked out.

Green's functions are computed analytically in the computer algebra system *Maple* imposing appropriate jump conditions that represent the Dirac delta in Eq. (8a). Their explicit reproduction is omitted here for the sake of brevity. The order of the variables in influence functions is according to Table 1. Consistent physical units are used.

5.1. Fixed 1d rod

A straight rod of length l, obeying 1d rod theory (Table 1), is given. The domain is $\Omega = (0, l)$. The rod is clamped at both ends. A non-constant axial stiffness $EA(x) = 5x+5$ is assumed. This setup is identical to the setup given in von Scheven et al. (2021).

The influence function for the displacement is obtained from Green's function, employing Eq. (12):

$$E_{\mathrm{u},\varepsilon^0}(x,s) = \mathrm{Heaviside}(x-s) - \frac{\ln(x+1)}{\ln(l+1)}, \quad x, s \in \Omega. \tag{36}$$

It describes the displacement distribution $u(x)$ in the structure caused by a displacement jump of 1 imposed at s.

The scalar-valued redundancy function is obtained by applying Eq. (15):

$$R(x,s) = -E_{\varepsilon^{\mathrm{el}},\varepsilon^0} = \frac{1}{\ln(l+1)(x+1)} = R(x), \quad x \in \Omega. \tag{37}$$

It describes the distribution of negative elastic strain $-\varepsilon^{\mathrm{el}}(x)$ in the structure caused by a displacement jump of 1 imposed at s.

The redundancy density according to Eq. (21) yields:

$$\bar{R}(x) = \frac{1}{\ln(l+1)(x+1)}, \quad x \in \Omega. \tag{38}$$

It describes the spatial distribution of the redundancy in the structure. Fig. 1 (left) shows the distribution of the redundancy density $\bar{R}(x)$ for $l = 10$. It is identical to the result in von Scheven et al. (2021) obtained by means of considering a truncated Taylor series expansion of the negative elastic elongation of an infinitesimal line element.

Redundancy matrices for $l = 10$ and for various discretizations assuming an equidistant grid of n_{e} elements ($e = 1, \dots, n_{\mathrm{e}}$) are computed according to Eq. (1). Fig. 1 (right) shows the main diagonal components of the corresponding redundancy matrices plotted as constant functions over x. It illustrates that the redundancy contribution of each element converges to the redundancy density (Fig. 1 (left)) at its location.

The trace of the redundancy operator using Eq. (23) is $\mathrm{tr}_\Omega(\mathcal{R}) = 1 = r$. The total redundancy r is equal to the degree of statical indeterminacy $n_{\mathrm{s}} = 1$ of the structure. It is induced by the support conditions.

The non-zero eigenvalue and corresponding normalized eigenfunction of the redundancy operator $\mathcal{R}$ according to Section 4.6 yield

$$\lambda = 1, \quad \phi(x) = \frac{1}{\ln(l+1)(x+1)}, \quad x \in \Omega. \tag{39}$$

The non-zero eigenvalue being equal to 1 confirms idempotence.

The self-stress function is obtained by Eq. (18):

$$S(x,s) = -EA(x)R(x,s) = E_{N,\varepsilon^0}(x,s) = -\frac{5}{\ln(l+1)} = \text{const.}, \quad x \in \Omega. \tag{40}$$

It describes the distribution of the normal force $N(x)$ in the structure caused by a displacement jump of 1 imposed at s.

The non-zero eigenvalue and corresponding normalized eigenfunction of the self-stress operator $\mathcal{S}$ according to Section 4.7 yield

$$\eta = -\frac{5l}{\ln(l+1)}, \quad \psi(x) = \frac{1}{l} = \text{const.} \tag{41}$$

Table 3
Subspaces of redundancy operator and total self-strain operator.

	Subspaces of $\mathcal{R}$ and $\mathcal{A}_0$ (dimension $\dim(\mathcal{R}) = \dim(\mathcal{A}_0) = \infty$)	
Subspace	$\mathrm{im}(\mathcal{R}) = \mathrm{null}(\mathcal{A}_0)$ Subspace of non-compatible initial strains	$\mathrm{null}(\mathcal{R}) = \mathrm{im}(\mathcal{A}_0)$ Subspace of compatible initial strains
Dimension	$\dim(\mathrm{im}(\mathcal{R})) = \dim(\mathrm{null}(\mathcal{A}_0))$ $= \mathrm{rank}(\mathcal{R}) = L = n_s$	$\dim(\mathrm{null}(\mathcal{R})) = \dim(\mathrm{im}(\mathcal{A}_0)) = \infty$
Initial strain	$\boldsymbol{\varepsilon}^0(x)$ such that $\int_\Omega \mathbf{R}(x,s)\boldsymbol{\varepsilon}^0(s)\,\mathrm{d}s = \boldsymbol{\varepsilon}^0(x)$ $\Leftrightarrow \boldsymbol{\varepsilon}^0(x) = \boldsymbol{\phi}(x),\ \lambda = 1$	$\boldsymbol{\varepsilon}^0(x)$ such that $\int_\Omega \mathbf{R}(x,s)\boldsymbol{\varepsilon}^0(s)\,\mathrm{d}s = \mathbf{0}$ $\Leftrightarrow \boldsymbol{\varepsilon}^0(x) = \boldsymbol{\phi}(x),\ \lambda = 0$
Mech. properties	- Stress-inducing, $\boldsymbol{\varepsilon}^{\mathrm{el}}(x) = \boldsymbol{\varepsilon}^0(x)$ - Deformation-free, $\boldsymbol{\epsilon}(x) = \mathbf{0} \Rightarrow \mathbf{u}(x) = \text{const.}$	- Stress-free, $\boldsymbol{\varepsilon}^{\mathrm{el}}(x) = \mathbf{0}$ - Deformation-inducing, $\boldsymbol{\epsilon}(x) = \boldsymbol{\varepsilon}^0(x)$

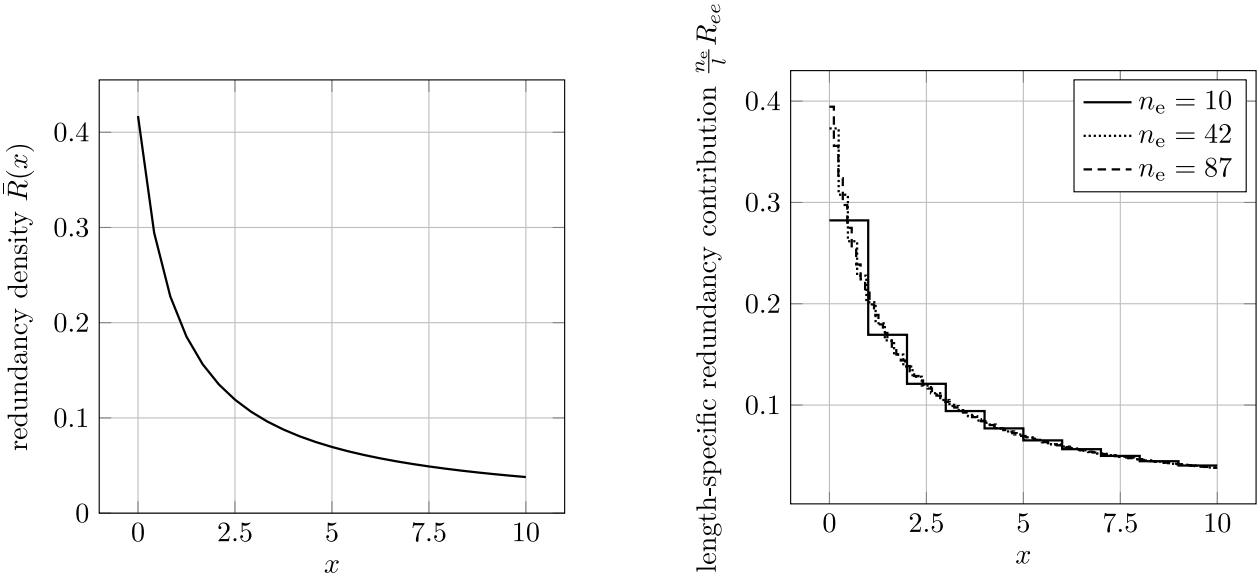


Fig. 1. Fixed 1d rod – redundancy density for $l = 10$ (left); main-diagonal components of redundancy matrices for various discretizations (right).

η may be considered as modal normal force quantity, ψ may be interpreted as initial axial strain. Thus, the product $\eta\psi$ may be interpreted as a normal force distribution that is equilibrated in the structure itself, i.e. in the absence of distributed external loads. $\eta\psi$ being constant is plausible, because it means a constant normal force in equilibrium.

5.2. Clamped 2d-Bresse-Timoshenko beam

A straight plane beam of length l, obeying 2d-Bresse-Timoshenko beam theory (Table 1), is considered. The domain is $\Omega = (0, l)$. The beam is clamped at both ends. To keep the resulting terms relatively simple, bending stiffness $EI = \text{const.}$ and shear stiffness $GA_S = \text{const.}$ are assumed.

The redundancy function is obtained by applying Eq. (15) (for $x, s \in \Omega$):

$$\begin{aligned}\mathbf{R}(x,s) &= \begin{bmatrix} -E_{\kappa^{\mathrm{el}},\kappa^0} & -E_{\kappa^{\mathrm{el}},\gamma^0} \\ -E_{\gamma^{\mathrm{el}},\kappa^0} & -E_{\gamma^{\mathrm{el}},\gamma^0} \end{bmatrix} = \begin{bmatrix} R_{\kappa\kappa} & R_{\kappa\gamma} \\ R_{\gamma\kappa} & R_{\gamma\gamma} \end{bmatrix} \\ &= -\frac{1}{l(GA_S l^2 + 12EI)} \\ &\quad \begin{bmatrix} (-4l^2 + (6s+6x)l - 12sx)GA_S - 12EI & 6GA_S(l-2x) \\ 6EI(l-2s) & -12EI \end{bmatrix}. \end{aligned} \tag{42}$$

For example, redundancy function component $R_{\kappa\kappa}(x, s)$ describes the distribution of the negative elastic curvature $-\kappa^{\mathrm{el}}(x)$ in the structure caused by a jump in the cross-sectional rotation of 1 imposed at s. A plot of $R_{\kappa\kappa}(x, s)$ for $l = 10$, $EI = 10$, and $GA_S = 1$ is shown in Fig. 2.

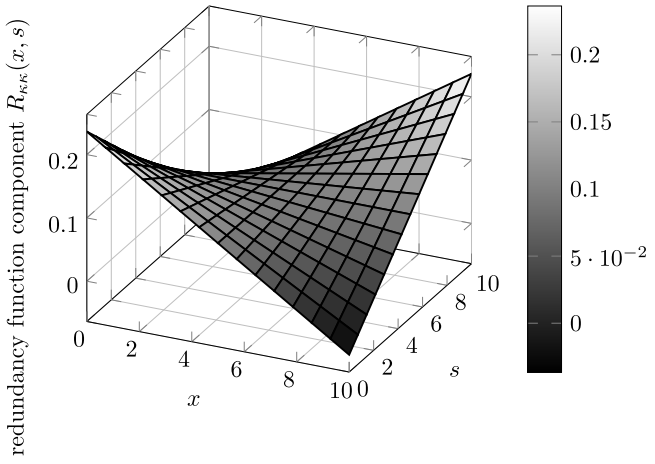


Fig. 2. Clamped 2d-Bresse-Timoshenko beam – redundancy function component $R_{\kappa\kappa}(x, s)$ for $l = 10$, $EI = 10$, and $GA_S = 1$.

The redundancy densities for curvature $\bar{R}_{\kappa\kappa}$ and for shear $\bar{R}_{\gamma\gamma}$ according to Eq. (21) yield

$$\bar{R}_{\kappa\kappa}(x) = \frac{(4l^2 - 12lx + 12x^2)GA_S + 12EI}{l(GA_S l^2 + 12EI)}, \quad x \in \Omega, \tag{43a}$$

$$\bar{R}_{\gamma\gamma}(x) = \frac{12EI}{l(GA_S l^2 + 12EI)} = \text{const.} \tag{43b}$$

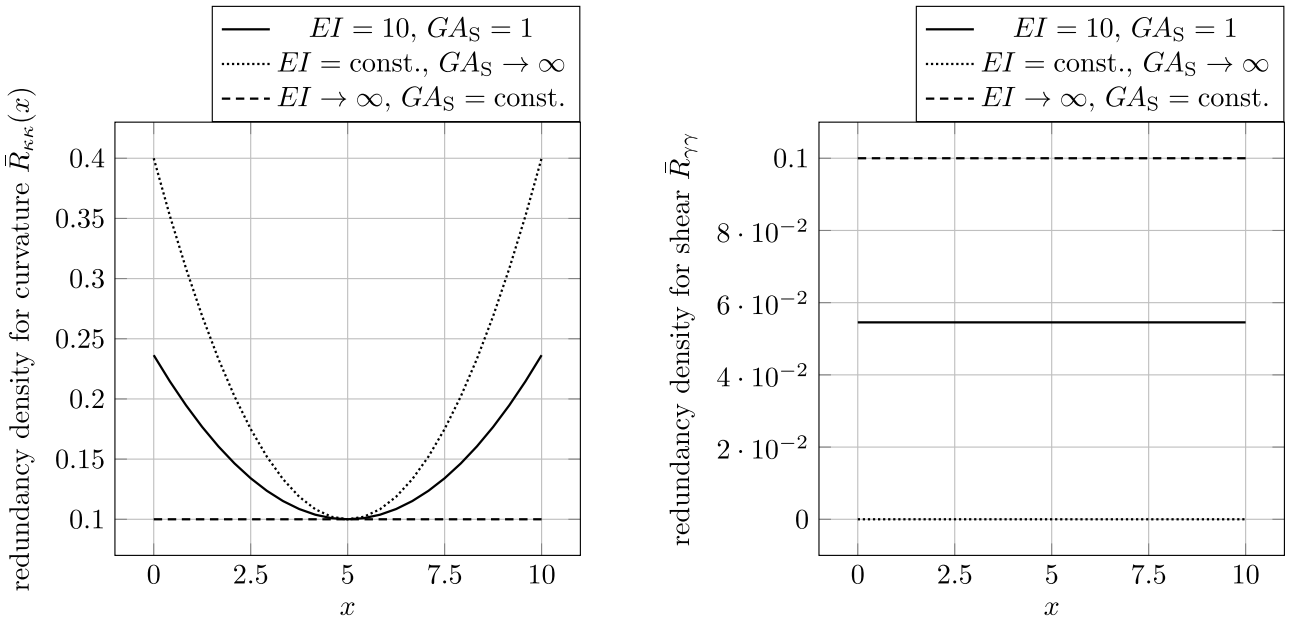


Fig. 3. Clamped 2d-Bresse-Timoshenko beam – redundancy densities for $l = 10$, various EI, and various GA_S.

The trace of the redundancy operator by using Eq. (23) yields:

$$\operatorname{tr}_\Omega(\mathcal{R}) = \sum_{i=\{\kappa,\gamma\}} \int_0^l \bar{R}_{ii}(x)\,\mathrm{d}x = \underbrace{\frac{2GA_S l^2 + 12EI}{GA_S l^2 + 12EI}}_{=r_{\kappa\kappa}} + \underbrace{\frac{12EI}{GA_S l^2 + 12EI}}_{=r_{\gamma\gamma}} = 2 = r.$$

$r_{\kappa\kappa}$ and $r_{\gamma\gamma}$ are the contributions of curvature and shear, respectively, to the redundancy in the structure, whereas the redundancy densities $\bar{R}_{\kappa\kappa}$ and $\bar{R}_{\gamma\gamma}$ give their spatial distributions. Plots of redundancy densities for $l = 10$, various EI, and various GA_S are shown in Fig. 3. The specific numerical values are not inspired by real beam problems but are meant to study the limit of infinite shear stiffness and bending stiffness, respectively. As the redundancy is induced by the clamping, it is plausible that the redundancy density for curvature increases towards the ends (for finite EI). The total redundancy of the structure r is equal to its degree of statical indeterminacy $n_s = 2$. It is induced by the support conditions.

The redundancy densities and redundancy contributions for infinite shear stiffness, i.e. $GA_S \to \infty$, are

$$\lim_{GA_S\to\infty} \bar{R}_{\kappa\kappa}(x) \to \frac{4l^2 - 12lx + 12x^2}{l^3}, \quad x \in \Omega \tag{44a}$$

$$\lim_{GA_S\to\infty} \bar{R}_{\gamma\gamma} \to 0, \tag{44b}$$

$$\lim_{GA_S\to\infty} r_{\kappa\kappa} \to 2, \tag{44c}$$

$$\lim_{GA_S\to\infty} r_{\gamma\gamma} \to 0. \tag{44d}$$

In this limit case, the entire redundancy is contributed by curvature (see also Fig. 3), which means that the structure effectively behaves according to the assumptions of an Euler-Bernoulli beam model. For $GA_S \to \infty$ (Eqs. (44)), the boundary conditions and the jump conditions $[[\varphi(s)]] = 0$, $[[w_S(s)]] = 1$ can be satisfied with vanishing $\bar{R}_{\gamma\gamma}$ (thus vanishing $r_{\gamma\gamma}$) because the kinematic equation defining the shear angle $\gamma(x) = \varphi(x) + \mathrm{d}w(x)/\mathrm{d}x$ depends on two unknown functions, whereas the curvature $\kappa(x) = \mathrm{d}\varphi(x)/\mathrm{d}x$ depends on one single unknown function.

The non-zero eigenvalues and corresponding normalized eigenfunctions of the redundancy operator $\mathcal{R}$ according to Section 4.6 are (for $x \in \Omega$):

$$\lambda_1 = 1,\ \boldsymbol{\phi}_1(x) = \frac{\bar{\boldsymbol{\phi}}_1(x)}{\|\bar{\boldsymbol{\phi}}_1(x)\|_2},\ \bar{\boldsymbol{\phi}}_1(x) = \begin{bmatrix} -GA_S(l-2x)/(2lEI) \\ 1/l \end{bmatrix}, \tag{45a}$$

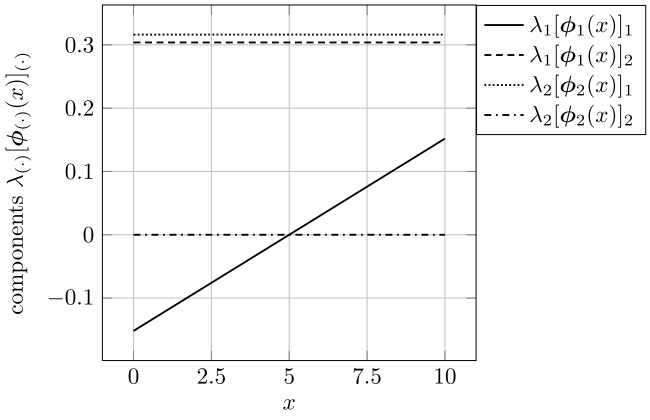


Fig. 4. Clamped 2d-Bresse-Timoshenko beam – $\lambda_{(\cdot)}\boldsymbol{\phi}_{(\cdot)}(x)$ for $l = 10$, $EI = 10$ and $GA_S = 1$.

$$\lambda_2 = 1,\ \boldsymbol{\phi}_2(x) = \frac{\bar{\boldsymbol{\phi}}_2(x)}{\|\bar{\boldsymbol{\phi}}_2(x)\|_2},\ \bar{\boldsymbol{\phi}}_2(x) = \begin{bmatrix} 1/l \\ 0 \end{bmatrix} = \text{const.} \tag{45b}$$

The two non-zero eigenvalues equal to 1 confirm idempotence. The eigenvalues and corresponding normalized eigenfunctions can be considered as modal redundancy quantities and independent distributions of initial strain quantities, respectively. Thus, the products $\lambda_{(\cdot)}\boldsymbol{\phi}_{(\cdot)}(x)$ can be interpreted as independent distributions of negative elastic self-strain quantities. Plots for $l = 10$, $EI = 10$ and $GA_S = 1$ are shown in Fig. 4. $\lambda_1[\boldsymbol{\phi}_1(x)]_1$ is the linear elastic curvature mode with constant elastic shear mode $\lambda_1[\boldsymbol{\phi}_1(x)]_2$, and $\lambda_2[\boldsymbol{\phi}_2(x)]_1$ is the constant elastic curvature mode with vanishing elastic shear mode $\lambda_2[\boldsymbol{\phi}_2(x)]_2$.

The self-stress function is obtained by Eq. (18) (for $x, s \in \Omega$):

$$\mathbf{S}(x,s) = -\mathbf{CR}(x,s) = \begin{bmatrix} E_{\mathrm{M},\kappa^0} & E_{\mathrm{M},\gamma^0} \\ E_{\mathrm{V},\kappa^0} & E_{\mathrm{V},\gamma^0} \end{bmatrix} = \mathbf{S}^{\mathrm{T}}(s,x)$$
$$= -\frac{1}{l(GA_S l^2 + 12EI)} \begin{bmatrix} EI((-4l^2 + (6s+6x)l - 12sx)GA_S - 12EI) & 6GA_S(l-2x)EI \\ \text{sym} & -12GA_S EI \end{bmatrix}.$$

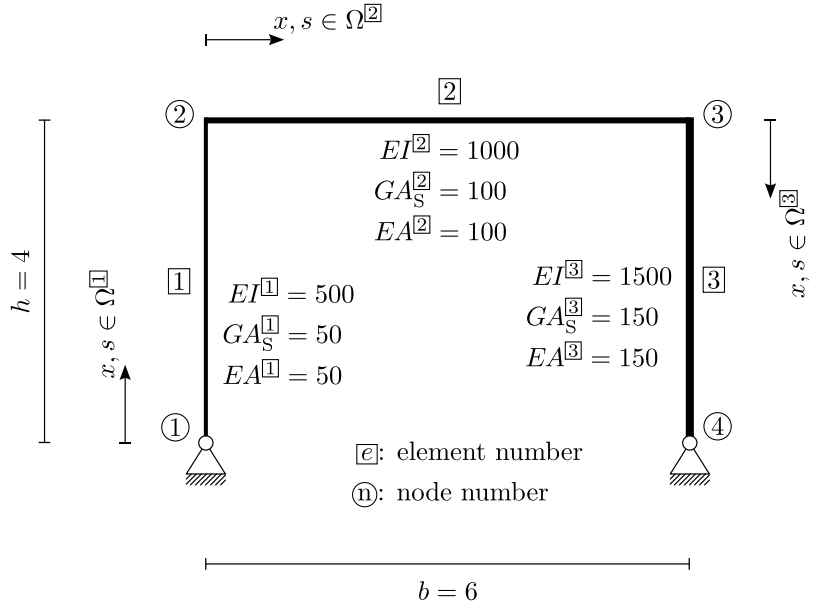


Fig. 5. Plane two-hinged frame 1 – system.

For example, $S_{12}(x, s)$ describes the distribution of the bending moment $M(x)$ in the structure caused by a jump of 1 in the shear displacement imposed at s. Symmetry with respect to Eq. (20) is confirmed.

The non-zero eigenvalues and corresponding normalized eigenfunctions of the self-stress operator $\mathcal{S}$ according to Section 4.7 are (for $x \in \Omega$):

$$\eta_1 = \frac{GA_S EI(l^2+12)}{GA_S l^2 + 12EI}, \quad \boldsymbol{\psi}_1(x) = \frac{\bar{\boldsymbol{\psi}}_1(x)}{\|\bar{\boldsymbol{\psi}}_1(x)\|_2}, \quad \bar{\boldsymbol{\psi}}_1(x) = \begin{bmatrix} -\frac{GA_S EI(l^2+12)(l-2x)}{2l(GA_S l^2+12EI)} \\ \frac{GA_S EI(l^2+12)}{l(GA_S l^2+12EI)} \end{bmatrix}, \tag{46a}$$

$$\eta_2 = EI, \qquad \boldsymbol{\psi}_2(x) = \frac{\bar{\boldsymbol{\psi}}_2(x)}{\|\bar{\boldsymbol{\psi}}_2(x)\|_2}, \quad \bar{\boldsymbol{\psi}}_2(x) = \begin{bmatrix} \frac{EI}{l} \\ 0 \end{bmatrix} = \text{const.} \tag{46b}$$

The normalized eigenfunctions form an orthonormal basis of the self-stress operator. The eigenvalues and corresponding normalized eigenfunctions can be considered as modal self-stress quantities and orthonormal distributions of initial strain quantities, respectively. Thus, the products $\eta_{(\cdot)}\boldsymbol{\psi}_{(\cdot)}(x)$ can be interpreted as orthonormal self-stress distributions. $\eta_{(\cdot)}[\boldsymbol{\psi}_{(\cdot)}(x)]_1$ represent the bending moments and $\eta_{(\cdot)}[\boldsymbol{\psi}_{(\cdot)}(x)]_2$ represent the shear forces.

5.3. Plane two-hinged frame 1

A plane frame of dimensions $b = 6$ and $h = 4$, obeying 2d-Bresse-Timoshenko beam theory for the curvature and shear parts as well as 1d rod theory for the axial strain part (Table 1), is considered. It is illustrated in Fig. 5. The domain is $\Omega = \Omega^{[1]} \cap \Omega^{[2]} \cap \Omega^{[3]}$, where $\Omega^{[1]} = (0, h)$, $\Omega^{[2]} = (h, h+b)$, $\Omega^{[3]} = (h+b, h+b+h)$. To keep the resulting terms simple, constant bending stiffness $EI^{[e]} = \text{const.}$, shear stiffness $GA_S^{[e]} = \text{const.}$, axial stiffness $EA^{[e]} = \text{const.}$ with simple values, given in Fig. 5, are assumed (academic example). The values that are chosen for this academic example do not match the magnitude in real frame structure. However, this specific choice facilitates relatively simple analytic expressions. Moreover, this does not have an effect on the results discussed below.

The BVP (8) is solved separately for $\Omega^{[1]}$, $\Omega^{[2]}$, and $\Omega^{[3]}$. Afterwards, the integration constants are determined by means of boundary conditions, transition conditions between the subdomains and the jump conditions from Table 2.

The redundancy function is a subdomain-wise defined, matrix-valued function of dimension 2×2. Exemplarily, $R_{\kappa\kappa}$ is given:

$$R_{\kappa\kappa}(x,s) = -E_{\kappa^{\text{el}},\kappa^0}(x,s) = \frac{1}{719}\begin{cases} 9xs/2, & s \in \Omega^{[1]}, x \in \Omega^{[1]} \\ 9s, & s \in \Omega^{[1]}, x \in \Omega^{[2]} \\ -3xs/2 + 21s, & s \in \Omega^{[1]}, x \in \Omega^{[3]} \\ 18x, & s \in \Omega^{[2]}, x \in \Omega^{[1]} \\ 36, & s \in \Omega^{[2]}, x \in \Omega^{[2]} \\ 84 - 6x, & s \in \Omega^{[2]}, x \in \Omega^{[3]} \\ -9xs/2 + 63x, & s \in \Omega^{[3]}, x \in \Omega^{[1]} \\ 126 - 9s, & s \in \Omega^{[3]}, x \in \Omega^{[2]} \\ 294 + 3xs/2 - 21x - 21s, & s \in \Omega^{[3]}, x \in \Omega^{[3]} \end{cases}. \tag{47}$$

Plots of the redundancy densities $\bar{R}_{\kappa\kappa}$, $\bar{R}_{\gamma\gamma}$, and $\bar{R}_{\varepsilon\varepsilon}$ are shown in Fig. 6. They show the spatial distributions of curvature, shear, and axial strain, respectively, to the redundancy in the frame. Exemplarily, $\bar{R}_{\kappa\kappa}$ is given:

$$\bar{R}_{\kappa\kappa}(x) = \frac{3}{1438}\begin{cases} 3x^2 & \text{for } x \in \Omega^{[1]} \\ 24 & \text{for } x \in \Omega^{[2]} \\ (x-14)^2 & \text{for } x \in \Omega^{[3]} \end{cases}.$$

The trace of the redundancy operator is:

$$\text{tr}_\Omega(\mathcal{R}) = \sum_{i=\{\kappa,\gamma,\varepsilon\}} \sum_{e=1}^{3} \underbrace{\int_{\Omega^{[e]}} \bar{R}_{ii}(x)\,\mathrm{d}x}_{=: r_{ii}^{[e]}} = 1 = r. \tag{48}$$

$r_{ii}^{[e]}$ is the redundancy contributed by strain type i of element $[e]$. The distribution of redundancy contributions is shown in Table 4. The total redundancy of the frame r is equal to its degree of statical indeterminacy $n_s = 1$. It is induced by the support conditions. The results are discussed in the sequel.

Load carrying behavior

Considering $\bar{R}_{ii}$ and r_{ii}, the following statements concerning the load carrying behavior of the structure can be derived. The axial forces in the columns as well as the shear force in the girder are non-redundant as $r_{\varepsilon\varepsilon}^{[1]} = r_{\varepsilon\varepsilon}^{[3]} = 0$ and $r_{\gamma\gamma}^{[2]} = 0$. Thus, these internal force

Table 4
Plane two-hinged frame 1 – distribution of redundancy contributions (rc) to strain types and elements (rounded to the fifth digit).

	rc from left column $r_{ii}^{\boxed{1}}$	rc from girder $r_{ii}^{\boxed{2}}$	rc from right column $r_{ii}^{\boxed{3}}$	Sums of rc $r_{ii} = \sum_{e=1}^{3} r_{ii}^{\boxed{e}}$
rc from curvature $r_{\kappa\kappa}^{\boxed{e}}$	0.13352	0.30042	0.044506	$0.47844 = r_{\kappa\kappa}$
rc from shear $r_{\gamma\gamma}^{\boxed{e}}$	0.25035	0	0.083449	$0.33380 = r_{\gamma\gamma}$
rc from axial strain $r_{\varepsilon\varepsilon}^{\boxed{e}}$	0	0.18776	0	$0.18776 = r_{\varepsilon\varepsilon}$
Sums of rc $r^{\boxed{e}} = \sum_{i=\{\kappa,\gamma,\varepsilon\}} r_{ii}^{\boxed{e}}$	0.38387	0.48818	0.12796	$1 = r = n_s$

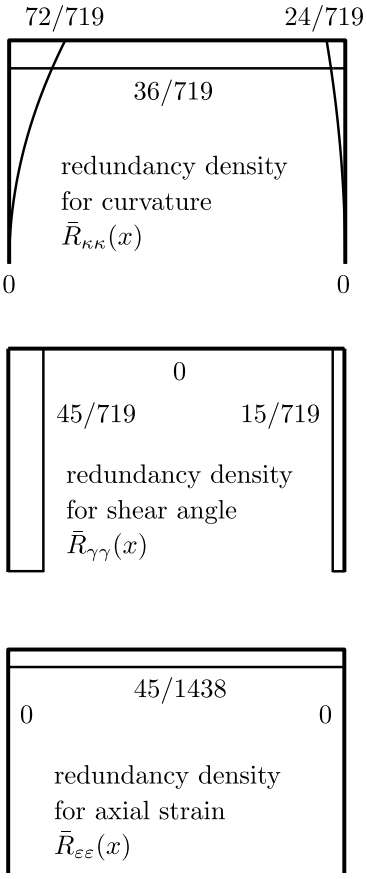


Fig. 6. Plane two-hinged frame 1 – redundancy densities (differently scaled).

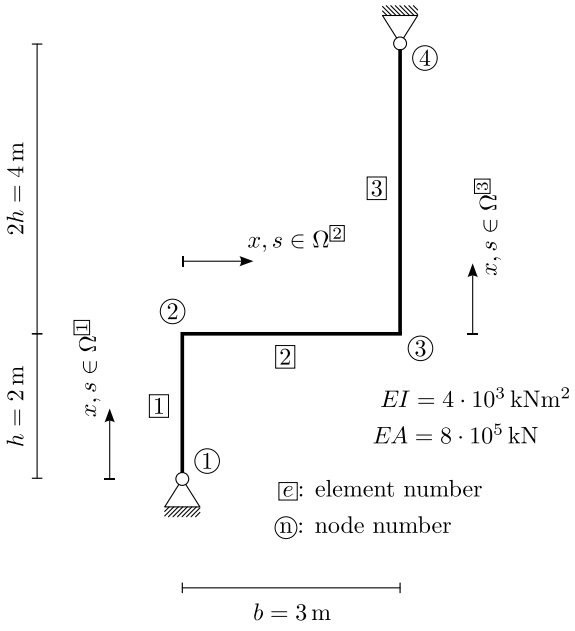


Fig. 7. Plane two-hinged frame 2 – system.

quantities may be computed from equilibrium equations only; these quantities are statically determinate and their magnitude is not affected by the stiffness distribution within the structure. Moreover, any constraint load case, e.g. temperature change or prescribed displacements at the supports, will neither induce normal forces in the columns nor transverse shear forces in the girder. Obviously, assuming an arbitrary load case, e.g. the removal of the axial stiffness in one column (by e.g. integrating an axial hinge) would lead to structural failure due to a violation of equilibrium. Conversely, the removal of the axial stiffness in the girder or the removal of the shear stiffness in one column would lead to an admissible statically determinate structure, fulfilling equilibrium equations.

Internal constraint

Redundancy densities $\bar{R}_{ii}$ as well as redundancy contributions r_{ii} provide information about distribution of internal constraint within the frame. For example, $r_{\varepsilon\varepsilon}^{\boxed{1}} = 0 \Leftrightarrow \bar{R}_{\varepsilon\varepsilon}(x) = 0$, $x \in \Omega^{\boxed{1}} \Leftrightarrow R_{\varepsilon\varepsilon}(x,s) = 0$, $x \in \Omega$, $s \in \Omega^{\boxed{1}}$ reflects the fact that imposing any axial elongation in the left column (by e.g. temperature or actuation) produces no elastic axial strain (i.e. no axial force) in the entire frame.

Furthermore, the off-diagonal components of the redundancy quantities ($i \neq k$) provide information about coupling of strain types i and k within the frame due to internal constraints. While $R_{ik}(x,s)$ provides this information also for the case of source and observation point not being identical, $\bar{R}_{ik}(x)$ presumes the locations of source and observation to be identical. Besides, $r_{ik}^{\boxed{e}}$ and r_{ik}, respectively, provide scalar measures for the coupling of different strain components within element e and for the entire frame, respectively, also presuming identical source and observation point. For example, $R_{\gamma\varepsilon}(x,s) = 0$, $x \in \Omega^{\boxed{2}}, s \in \Omega^{\boxed{1}}$ reflects the fact that imposing a jump in the axial displacement in the left column (by e.g. concentrated actuation) produces no elastic shear angle (i.e. no shear force) in the girder. Conversely, $R_{\gamma\varepsilon}(x,s) \neq 0$, $x \in \Omega^{\boxed{1}}$, $s \in \Omega^{\boxed{2}}$ means that imposing a jump in the axial displacement in the girder produces a non-zero elastic shear angle (i.e. non-zero shear force) in the left column. Moreover, $\bar{R}_{\kappa\varepsilon}(x) \neq 0$, $x \in \Omega^{\boxed{2}} \Leftrightarrow r_{\kappa\varepsilon}^{\boxed{2}} \neq 0$ exemplarily reflects the fact that imposing a jump in the axial displacement in the girder produces non-zero elastic curvature (i.e. non-zero bending moment) at the point of imposition.

Assessment of adaptability

The eigenfunction $\boldsymbol{\phi}(x)$ corresponding to the single non-zero eigenvalue (omitted here) spans $\mathrm{im}(\mathcal{R})$. According to Table 3, it forms the one-dimensional function space of non-compatible, deformation-free self-stress states. These states can be induced by imposing initial strains (by means of e.g. actuation) and can be used to change states of stress imposed by external loads. For example, consider some external loads $\mathbf{p}(x)$ inducing the stress state $\boldsymbol{\sigma}(x)$ and assume the initial strains $\boldsymbol{\varepsilon}^0(x) = \alpha\,\boldsymbol{\phi}(x)$, the magnitude of which is controlled by the parameter $\alpha \in \mathbb{R}$. Then the resulting stress state is $\tilde{\boldsymbol{\sigma}}(x) = \boldsymbol{\sigma}(x) + \alpha \int_\Omega \mathbf{S}(x,s)\boldsymbol{\phi}(s)\,\mathrm{d}s$. In this light, the redundancy operator and the self-stress operator, especially their spectral properties, provide information on theoretical limits of adaptability of the frame.

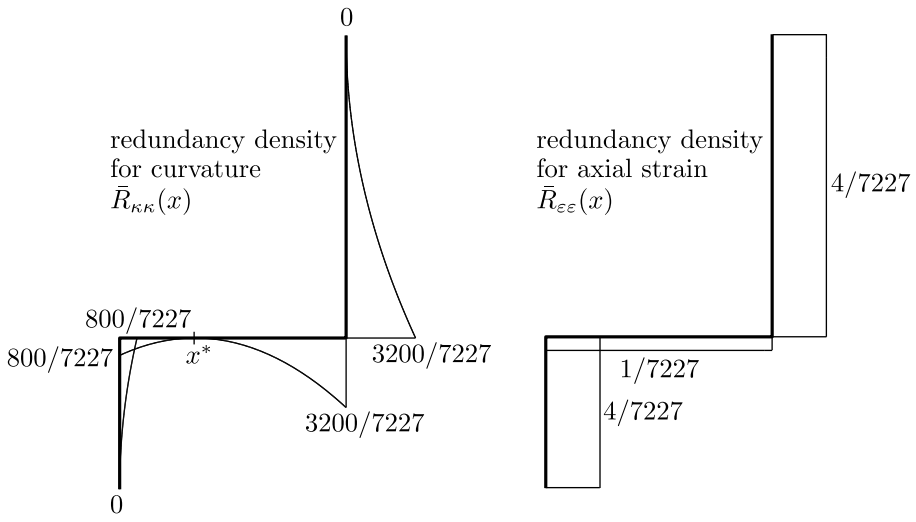


Fig. 8. Plane two-hinged frame 2 – redundancy densities (differently scaled).

5.4. Plane two-hinged frame 2

A plane frame of overall dimensions $b = 3\,\mathrm{m}$ and $3h = 6\,\mathrm{m}$, obeying 2d-Euler-Bernoulli beam theory for the curvature part as well as 1d rod theory for the axial strain part (Table 1), is given. It is illustrated in Fig. 7. The domain is $\Omega = \Omega^{\boxed{1}} \cap \Omega^{\boxed{2}} \cap \Omega^{\boxed{3}}$, where $\Omega^{\boxed{1}} = (0, h)$, $\Omega^{\boxed{2}} = (h, h+b)$, $\Omega^{\boxed{3}} = (h+b, h+b+2h)$. The bending stiffness $EI = \text{const.}$ as well as axial stiffness $EA = \text{const.}$ are assumed to be equal in all elements. They take on the values given in Fig. 7, which is equivalent to choosing the structural steel cross-section HEA 160 (rounded values).

Plots of the redundancy densities for curvature $\bar{R}_{\kappa\kappa}$ and axial strain $\bar{R}_{\varepsilon\varepsilon}$ are shown in Fig. 8. Exemplarily, $\bar{R}_{\kappa\kappa}$ is given:

$$\bar{R}_{\kappa\kappa}(x) = \frac{200}{7227}\frac{1}{\mathrm{m}}\begin{cases} x^2 & \text{for } x \in \Omega^{\boxed{1}} \\ 4(x-3)^2 & \text{for } x \in \Omega^{\boxed{2}} \\ (x-9)^2 & \text{for } x \in \Omega^{\boxed{3}} \end{cases}.$$

The trace of the redundancy operator are computed analogously to Eq. (48) and is $\mathrm{tr}_\Omega(\mathcal{R}) = 1 = r$. The distribution of redundancy contributions to strain types and elements is shown in Table 5. $r_{\kappa\kappa}$ and $r_{\varepsilon\varepsilon}$ are the contributions of curvature and axial strain, respectively, to the redundancy in the frame. The total redundancy of the frame r is equal to its degree of statical indeterminacy $n_s = 1$. It is induced by the support conditions. In the following, some implication of these results on the interpretation and assessment of the load carrying behavior are discussed that go beyond the ones already described in the context of the preceding example in Section 5.3.

Load carrying behavior

Firstly, the redundancy density for the curvature at $x^* = h + b/3$ (intersection of element $\boxed{2}$ with a straight line connecting nodes ① and ④) is $\bar{R}_{\kappa\kappa}(x^*) = 0$ (see also Fig. 8). This corresponds to the fact that the bending moment $M(x^*)$ may be computed from the equilibrium equations at this point, i.e. it is statically determinate. This can also be seen from the fact that in any admissible unit force state in the force method the lever arms of the reaction forces with respect to x^* vanish (and thus the corresponding bending moment is zero). Consequently, $M(x^*)$ is independent of the stiffness distribution in the frame. This reflects the fact that static indeterminacy in truss and frame structures is a property originally related to structural points and neither to structural members nor to the entire structure.

Secondly, a comparison of the redundancy contributions, i.e. $r_{\kappa\kappa} = 0.99626 \gg 0.0037360 = r_{\varepsilon\varepsilon}$ and $r^{\boxed{e}}_{\kappa\kappa} \gg r^{\boxed{e}}_{\varepsilon\varepsilon}$ reveals that for this frame curvature (bending) is much more redundant than axial strain (normal force) is. This can be explained by the fact that the given structure exhibits a bending dominated load carrying behavior due to its topology. In a truss-like topology, the opposite would be true.

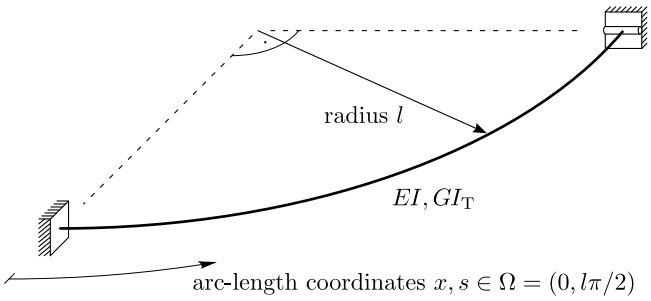


Fig. 9. Quarter-circular plane Euler-Bernoulli beam – system.

5.5. Quarter-circular plane Euler-Bernoulli beam

A quarter-circular plane beam with radius l, obeying the theory of a curvilinear plane Euler-Bernoulli beam under out-of-plane loading (Table 1), is given. The domain is $\Omega = (0, l\pi/2)$. As illustrated in Fig. 9, it is clamped at $x = 0$ and supported by a bending hinge at $x = l\pi/2$. To keep the resulting terms simple, constant bending stiffness $EI = \text{const.}$ and torque stiffness $GI_\mathrm{T} = \text{const.}$ are assumed.

The redundancy function can be computed. $R_{\vartheta\kappa}(x, s)$ is

$$R_{\vartheta\kappa}(x,s) = -E_{\vartheta^{\mathrm{el}},\kappa^0}(x,s) = \frac{4EI\cos(s/l)(\pi\sin(x/l) - 2)}{((GI_\mathrm{T}\pi^2 + EI(\pi^2 - 8))l)}, \quad x, s \in \Omega. \tag{49}$$

$R_{\vartheta\kappa}(x, s)$ describes the distribution of negative elastic drilling $-\vartheta^{\mathrm{el}}(x)$ in the structure caused by a jump of 1 in the cross-sectional rotation $\mathrm{d}w(s)/\mathrm{d}s$ imposed at s. $R_{\vartheta\kappa}$ being a function in x and s, reflects the fact, that bending and torque are coupled within this beam. Imposing a jump in cross-sectional rotation produces non-zero elastic drilling and thus a non-zero torque moment in the entire beam, except at $x^* = l\arcsin(2/\pi)$ as $R_{\vartheta\kappa}(x^*, s) = 0$.

The redundancy densities for curvature $\bar{R}_{\kappa\kappa}$ and drilling $\bar{R}_{\vartheta\vartheta}$ are

$$\bar{R}_{\kappa\kappa}(x) = \frac{4\cos(x/l)^2\pi GI_\mathrm{T}}{(l((EI + GI_\mathrm{T})\pi^2 - 8EI))}, \quad x \in \Omega, \tag{50a}$$

$$\bar{R}_{\vartheta\vartheta}(x) = \frac{(4EI\pi\sin(x/l)^2 - 16EI\sin(x/l) + 2(EI + GI_\mathrm{T})\pi)}{((EI\pi^2 + GI_\mathrm{T}\pi^2 - 8EI)l)}, \quad x \in \Omega. \tag{50b}$$

Table 5
Plane two-hinged frame 2 – distribution of redundancy contributions (rc) to strain types and elements (rounded to the fifth digit).

	rc from left column $r_{ii}^{\boxed{1}}$	rc from girder $r_{ii}^{\boxed{2}}$	rc from right column $r_{ii}^{\boxed{3}}$	Sums of rc $r_{ii} = \sum_{e=1}^{3} r_{ii}^{\boxed{e}}$
rc from curvature $r_{\kappa\kappa}^{\boxed{e}}$	0.073797	0.33209	0.59038	$0.99626 = r_{\kappa\kappa}$
rc from axial strain $r_{\varepsilon\varepsilon}^{\boxed{e}}$	0.0011069	0.00041511	0.0022139	$0.0037360 = r_{\varepsilon\varepsilon}$
Sums of rc $r^{\boxed{e}} = \sum_{i=\{\kappa,\varepsilon\}} r_{ii}^{\boxed{e}}$	0.074904	0.33250	0.59259	$1 = r = n_s$

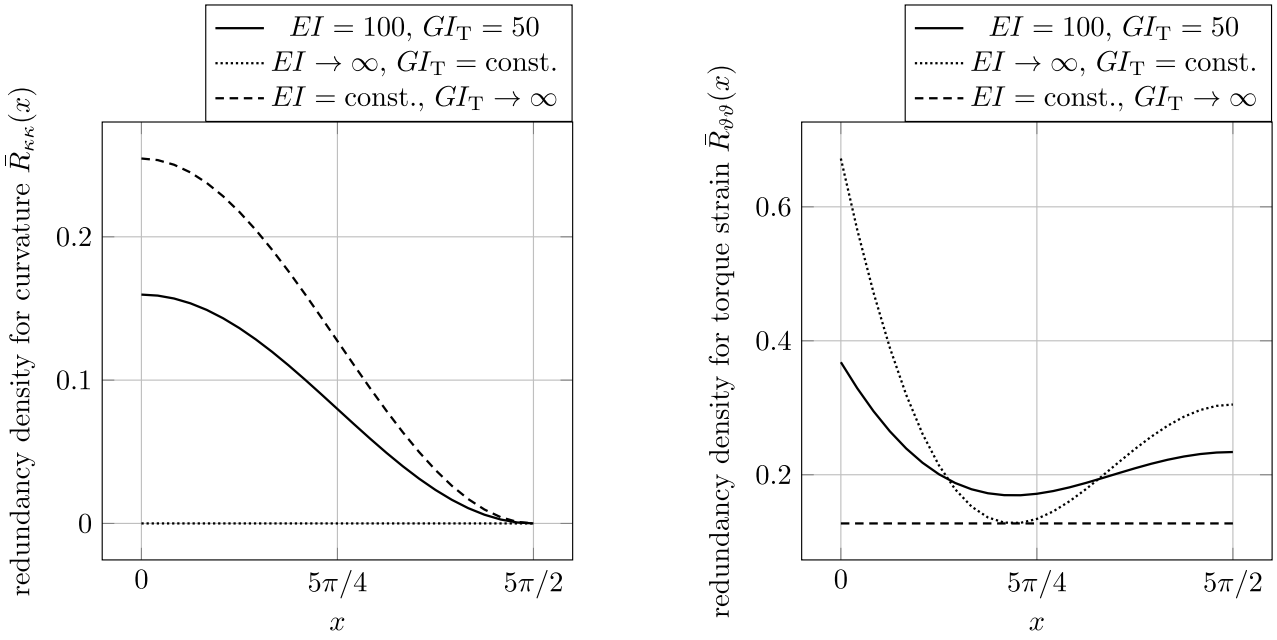


Fig. 10. Quarter-circular plane Euler-Bernoulli beam – redundancy densities for $l = 5$, various EI, and various GI_T (differently scaled).

The trace of the redundancy operator is

$$\mathrm{tr}_\Omega(\mathcal{R}) = \underbrace{\frac{\pi^2 GI_T}{((EI + GI_T)\pi^2 - 8EI)}}_{=r_{\kappa\kappa}} + \underbrace{\frac{(\pi^2(2EI + GI_T) - 16EI)}{((EI + GI_T)\pi^2 - 8EI)}}_{=r_{\vartheta\vartheta}} = 2 = r. \quad (51)$$

$r_{\kappa\kappa}$ and $r_{\vartheta\vartheta}$ are the contributions of curvature and drilling, respectively, to structural redundancy. The redundancy densities $\bar{R}_{\kappa\kappa}$ and $\bar{R}_{\vartheta\vartheta}$ quantify their spatial distributions. Plots of redundancy densities for $l = 5$, various EI, and various GI_T (academic example) are shown in Fig. 10. The redundancy associated with the drilling $r_{\vartheta\vartheta}$ is induced by redundant support conditions w.r.t. out-of-plane displacement and torque rotation, whereas the support situation with respect to cross-sectional rotation is non-redundant. Thus, it is plausible that the redundancy density $\bar{R}_{\vartheta\vartheta}$ increases towards the supports. Furthermore, the non-redundant support situation with respect to cross-sectional rotation is reflected by $\bar{R}_{\kappa\kappa}(l\pi/2) = 0$. The redundancy related to curvature $r_{\kappa\kappa}$ is induced by one redundant support of vertical displacement, whereas support of cross-sectional rotation is non-redundant. The total redundancy of the structure r is equal to its degree of statical indeterminacy $n_s = 2$.

6. Conclusion

The main result of the paper is the extension of the concept of discrete *redundancy matrices* to *redundancy operators* in continuous representations of structures. These are integral operators, related to Green's functions in elastostatics, that map initial strain quantities onto negative elastic strain quantities. Their integral kernel, denoted here as redundancy function, is a particular instance of an influence function. Furthermore, the *total self-strain operator* and the relation between its integral kernel, the total self-strain function, and the redundancy function have been derived.

Both operators are idempotent and their sum equals the identity operator. Thus, they share the same eigenfunctions and their eigenvalues are either 1 or 0. While the number of its eigenvalues equal to zero is infinite, the number of eigenvalues equal to 1 is finite and coincides with the degree of statical indeterminacy of the structure. The *redundancy density* and the trace properties of the redundancy operator provide information about the distribution of statical indeterminacy in space and with respect to the different load carrying types. This provides valuable insight into structural behavior that reaches far beyond standard interpretation of statical indeterminacy as one aggregated integer number.

A mechanical interpretation of the operators' subspaces reproduces a known decomposition of the space of initial strain quantities into two orthogonal subspaces: Firstly, the subspace of non-compatible initial strains, inducing stress but no deformation, and secondly, the subspace of compatible initial strains inducing deformation but no stress.

A further theoretical development is the introduction of the self-adjoint *self-stress operator*. Its normalized eigenfunctions may be interpreted as distributions of initial strain quantities, with the corresponding eigenvalues being modal self-stress quantities. The products of the corresponding eigenfunctions and eigenvalues yield orthonormal self-stress distributions.

The theoretical findings have been validated and exemplarily demonstrated and illustrated for various simple structures. For the example of a clamped 2d-Bresse-Timoshenko beam structure the split of the redundancy densities and contributions to the load carrying mechanisms into contributions from bending and shear have been illustrated. The limit case of infinite shear stiffness reproduces structural behavior according to Euler-Bernoulli beam theory, which validates the theoretical results. Further examples illustrate the potential of the theory to assist analysis and interpretation of load carrying behavior, for instance regarding the distribution of internal constraints and the potential for manipulating internal forces in adaptive structures.

Possible directions of future work include generalization of continuum-mechanical theory of redundancy to structural theories with internal statical indeterminacy (e.g. 2d- and 3d-continua) as well as for non-linear problems and dynamics.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

A.1. Proof of symmetry of Green’s function for linear elastostatics

The following assumptions are introduced:

- Linear theory is considered, thus superposition principle is valid.
- Homogeneous boundary conditions are assumed.
- Different external load distributions, $\mathbf{b}_\mathrm{P}$ and $\mathbf{b}_\mathrm{Q}$, are considered.
- No initial strain loads are imposed, i.e. $\boldsymbol{\varepsilon}^0 = \mathbf{0}$.

External work done by imposed external load distribution $\mathbf{b}_\mathrm{P}$:

$$W_\mathrm{P}^\mathrm{ext} = -\frac{1}{2}\int_\Omega \mathbf{u}_\mathrm{P}{}^\mathrm{T}(x)\mathbf{b}_\mathrm{P}(x)\,\mathrm{d}x \tag{52}$$

with

$$\mathbf{u}_\mathrm{P}(x) = \int_\Omega \mathbf{G}(x,s)\mathbf{b}_\mathrm{P}(s)\,\mathrm{d}s. \tag{53}$$

Applying Betti’s reciprocity theorem, inserting Eq. (53), transposing right-hand side, and changing variables yield:

$$\begin{aligned} W_\mathrm{PQ}^\mathrm{ext} &= W_\mathrm{QP}^\mathrm{ext} \\ \int_\Omega \mathbf{u}_\mathrm{P}{}^\mathrm{T}(x)\mathbf{b}_\mathrm{Q}(x)\,\mathrm{d}x &= \int_\Omega \mathbf{u}_\mathrm{Q}{}^\mathrm{T}(x)\mathbf{b}_\mathrm{P}(x)\,\mathrm{d}x \\ \int_\Omega\int_\Omega \mathbf{b}_\mathrm{P}{}^\mathrm{T}(s)\mathbf{G}^\mathrm{T}(x,s)\mathbf{b}_\mathrm{Q}(x)\,\mathrm{d}s\,\mathrm{d}x &= \int_\Omega\int_\Omega \mathbf{b}_\mathrm{Q}{}^\mathrm{T}(s)\mathbf{G}^\mathrm{T}(x,s)\mathbf{b}_\mathrm{P}(x)\,\mathrm{d}x \\ &= \int_\Omega\int_\Omega \mathbf{b}_\mathrm{P}{}^\mathrm{T}(s)\mathbf{G}(s,x)\mathbf{b}_\mathrm{Q}(x)\,\mathrm{d}s\,\mathrm{d}x. \end{aligned} \tag{54}$$

Considering fundamental lemma of variational calculus, Eq. (54) holds iff

$$G_{bn}(x,s) = G_{nb}(s,x). \tag{55}$$

Using Eq. (8b) and the symmetry property of δ yield:

$$\mathcal{D}^\mathrm{x}_{mn}G_{nb}(x,s) = \tilde{\delta}_{mb}\delta(s-x) = \tilde{\delta}_{mb}\delta(x-s) = \mathcal{D}^\mathrm{x}_{mn}G_{nb}(s,x) \tag{56}$$

Applying the derivation of the solution of BVP (7a) as an integral (analogously to Eq. (9)) to the leftmost and rightmost side of Eq. (56), it yields

$$u_n(x) = \int_\Omega G_{nb}(x,s)p_b(s)\,\mathrm{d}s = \int_\Omega G_{nb}(s,x)p_b(s)\,\mathrm{d}s. \tag{57}$$

Considering fundamental lemma of variational calculus, Eq. (57) holds, iff

$$G_{nb}(x,s) = G_{nb}(s,x). \tag{58}$$

Inserting Eq. (10a) into Eq. (10b) and changing variables proves Eq. (10c) (transitivity).

A.2. Proof of idempotence of redundancy and total self-strain operator

The indices are valid only for the present appendix.

The total self-strain operator is defined by Eq. (13). Considering an arbitrary sufficiently continuously-differentiable function $\mathbf{a}(x)$, the idempotence of the total self-strain operator is proven as follows:

$$\begin{aligned} \mathcal{A}_{0ik}\mathcal{A}_{0kf}a_f(x) &= \int_\Omega\int_\Omega A_{0ik}(x,z)A_{0kf}(z,s)\,\mathrm{d}z\,a_f(s)\,\mathrm{d}s \\ &= \int_\Omega\int_\Omega \Big(\mathcal{L}^\mathrm{x}_{in}C_{pk}(z)\mathcal{L}^z_{pb}G_{nb}(x,z)\Big)\Big(\mathcal{L}^z_{kq}C_{jf}(s)\mathcal{L}^\mathrm{s}_{jv}G_{qv}(z,s)\Big)\,\mathrm{d}z\,a_f(s)\,\mathrm{d}s \\ &= \int_\Omega\int_\Omega \Big(\mathcal{L}^\mathrm{x}_{in}\underbrace{\underbrace{\mathcal{L}^{z*}_{qk}C_{pk}(z)\mathcal{L}^z_{pb}}_{=D^z_{qb}}G_{nb}(x,z)}_{=\tilde{\delta}_{qn}\delta(z-x)}\Big)\Big(C_{jf}(s)\mathcal{L}^\mathrm{s}_{jv}G_{qv}(z,s)\Big)\,\mathrm{d}z\,a_f(s)\,\mathrm{d}s + \cancel{\mathrm{BT}} \\ &= \int_\Omega\int_\Omega \mathcal{L}^\mathrm{x}_{in}\tilde{\delta}_{qn}\delta(z-x)\Big(C_{jf}(s)\mathcal{L}^\mathrm{s}_{jv}G_{qv}(z,s)\Big)\,\mathrm{d}z\,a_f(s)\,\mathrm{d}s \\ &= \int_\Omega \Big(\mathcal{L}^\mathrm{x}_{iq}C_{jf}(s)\mathcal{L}^\mathrm{s}_{jv}G_{qv}(x,s)\Big)\,a_f(s)\,\mathrm{d}s \\ &= \int_\Omega A_{0if}(x,s)a_f(s)\,\mathrm{d}s = \mathcal{A}_{0if}a_f(s) \\ \Leftrightarrow\quad & \mathcal{A}_{0ik}\mathcal{A}_{0kf} = \mathcal{A}_{0if}, \quad \forall\, i,k,f. \end{aligned} \tag{59}$$

The boundary terms (BT) vanish as boundary conditions for $G_{(\cdot)}$ in x and z, respectively, are equivalent for an identical structure.

Along with the idempotence of the identity operator and relation Eq. (17), the idempotence of the redundancy operator is obvious.

A.3. Proof of symmetry properties of self-stress function

The following assumptions are introduced:

- Linear theory is considered, thus superposition principle is valid.
- Homogeneous boundary conditions are assumed.
- Different initial strain distributions, $\boldsymbol{\varepsilon}^0_\mathrm{P}$ and $\boldsymbol{\varepsilon}^0_\mathrm{Q}$, are considered as loads.
- No external loads are imposed, i.e. $\mathbf{p} = \mathbf{0}$.

External work done by imposed initial strain distribution $\boldsymbol{\varepsilon}^0_\mathrm{P}$ (Fig. 11):

$$W_\mathrm{P}^\mathrm{ext} = -\frac{1}{2}\int_\Omega \boldsymbol{\sigma}_\mathrm{P}{}^\mathrm{T}(x)\boldsymbol{\varepsilon}^0_\mathrm{P}(x)\,\mathrm{d}x, \tag{60}$$

with

$$\boldsymbol{\sigma}_\mathrm{P}(x) = \int_\Omega \mathbf{S}(x,s)\boldsymbol{\varepsilon}^0_\mathrm{P}(s)\,\mathrm{d}s. \tag{61}$$

External work done by initial strain distribution $\boldsymbol{\varepsilon}^0_\mathrm{Q}$ superimposed on $\boldsymbol{\varepsilon}^0_\mathrm{P}$:

$$\Delta W_\mathrm{P+Q}^\mathrm{ext} = -\frac{1}{2}\int_\Omega \boldsymbol{\sigma}_\mathrm{Q}{}^\mathrm{T}(x)\boldsymbol{\varepsilon}^0_\mathrm{Q}(x)\,\mathrm{d}x - \int_\Omega \boldsymbol{\sigma}_\mathrm{P}{}^\mathrm{T}(x)\boldsymbol{\varepsilon}^0_\mathrm{Q}(x)\,\mathrm{d}x. \tag{62}$$

Total external work done:

$$W_\mathrm{P+Q}^\mathrm{ext} = W_\mathrm{P}^\mathrm{ext} + \Delta W_\mathrm{P+Q}^\mathrm{ext} \tag{63}$$

Reversing order of imposition of initial strain distributions $\boldsymbol{\varepsilon}^0_\mathrm{P}$ and $\boldsymbol{\varepsilon}^0_\mathrm{Q}$, respectively, yields the same state with equal external work $W_\mathrm{Q+P}^\mathrm{ext}$. Analogously to Eq. (54) $W_\mathrm{P+Q}^\mathrm{ext}$ and $W_\mathrm{Q+P}^\mathrm{ext}$ are compared, yielding:

$$\int_\Omega \boldsymbol{\sigma}_\mathrm{Q}{}^\mathrm{T}(x)\boldsymbol{\varepsilon}^0_\mathrm{P}(x)\,\mathrm{d}x = \int_\Omega \boldsymbol{\sigma}_\mathrm{P}{}^\mathrm{T}(x)\boldsymbol{\varepsilon}^0_\mathrm{Q}(x)\,\mathrm{d}x. \tag{64}$$

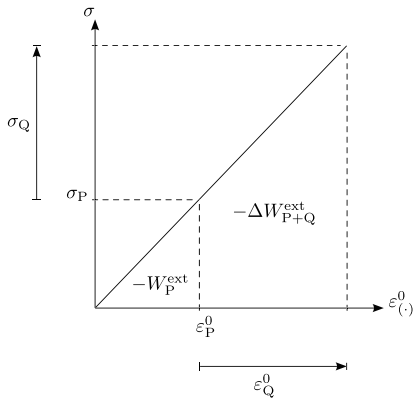


Fig. 11. Appendix A.3 – stress, strain and external work for 1d case.

Eq. (64) is Betti's reciprocity theorem for the case of prescribed initial strains from subspace $\mathrm{im}(S)$ as loads (a special case of which are strains due to thermal expansion). Inserting Eq. (61) into Eq. (64), changing variables on the left-hand side, and transposing right-hand side:

$$\int_\Omega \int_\Omega \boldsymbol{\varepsilon}_{\mathrm{Q}}^{0\mathrm{T}}(x)\mathbf{S}^{\mathrm{T}}(s,x)\boldsymbol{\varepsilon}_{\mathrm{P}}^{0}(s)\,\mathrm{d}s\,\mathrm{d}x = \int_\Omega \int_\Omega \boldsymbol{\varepsilon}_{\mathrm{Q}}^{0\mathrm{T}}(x)\mathbf{S}(x,s)\boldsymbol{\varepsilon}_{\mathrm{P}}^{0}(s)\,\mathrm{d}s\,\mathrm{d}x. \tag{65}$$

Considering fundamental lemma of variational calculus, Eq. (65) holds, iff

$$\mathbf{S}(x,s) = \mathbf{S}^{\mathrm{T}}(s,x). \tag{66}$$

A.4. Solution of eigenvalue equation for matrix-valued integral kernel

The symbols and indices are valid only for the present appendix.

Consider the integral operator $\mathcal{E} : C(\Omega,\mathbb{R}^N) \to C(\Omega,\mathbb{R}^N)$ defined by

$$\mathcal{E}\mathbf{b}(x) := \int_\Omega \mathbf{E}(x,s)\mathbf{b}(s)\,\mathrm{d}s, \tag{67}$$

with a separable, matrix-valued integral kernel $\mathbf{E} : \Omega^2 \to \mathbb{R}^{N\times N}$

$$\mathbf{E}(x,s) = \sum_{a=1}^{A} \mathbf{v}_a(x)\mathbf{q}_a{}^{\mathrm{T}}(s), \quad \mathbf{v}_a, \mathbf{q}_a : \Omega \to \mathbb{R}^N. \tag{68}$$

The corresponding vector-valued eigenvalue equation reads

$$\mathcal{E}\boldsymbol{\phi}(x) = \int_\Omega \mathbf{E}(x,s)\boldsymbol{\phi}(s)\,\mathrm{d}s = \lambda\boldsymbol{\phi}(x), \tag{69}$$

where λ is the eigenvalue and $\boldsymbol{\phi} : \Omega \to \mathbb{R}^N$ is the corresponding vector-valued eigenfunction of the integral operator. Replacing $\mathbf{E}$ in Eq. (69) yields

$$\lambda\boldsymbol{\phi}(x) = \sum_{a=1}^{A} \mathbf{v}_a(x) \int_\Omega \mathbf{q}_a{}^{\mathrm{T}}(s)\boldsymbol{\phi}(s)\,\mathrm{d}s. \tag{70}$$

A vector $\mathbf{c} \in \mathbb{R}^A$ is defined, the components of which are

$$c_p := \int_\Omega \mathbf{q}_p{}^{\mathrm{T}}(x)\boldsymbol{\phi}(x)\,\mathrm{d}x, \quad p = 1,\dots,A. \tag{71}$$

Multiplying Eq. (71) by λ, replacing $\lambda\boldsymbol{\phi}(s)$ by Eq. (70), and using Eq. (71), yields

$$\lambda c_p = \sum_{a=1}^{A} T_{pa}c_a = \lambda \sum_{a=1}^{A} \tilde{\delta}_{pa}c_a. \tag{72}$$

with the components of a quadratic matrix $\mathbf{T} \in \mathbb{R}^{A\times A}$ being

$$T_{pa} := \int_\Omega \mathbf{q}_p{}^{\mathrm{T}}(s)\mathbf{v}_a(s)\,\mathrm{d}s, \tag{73}$$

and $\tilde{\delta}_{pa}$ being Kronecker delta. Writing Eq. (72) in matrix notation, i.e. $\mathbf{T}\mathbf{c}_p = \lambda_p\mathbf{c}_p$, the eigenvalue equation (69) is traced back to a matrix eigenvalue problem. Its non-trivial solution yields A eigenvalues and the corresponding eigenvectors. Considering Eqs. (70) and (71), the pth vector-valued eigenfunction of the integral kernel reads

$$\boldsymbol{\phi}_p(x) = d_p \sum_{a=1}^{A} \mathbf{v}_a(x)[\mathbf{c}_p]_a, \tag{74}$$

where $[\mathbf{c}_p]_a$ is the ath component of the pth eigenvector and d_p is the pth constant, determinable by the L^2 normalization condition $\|\boldsymbol{\phi}_p(x)\|_2 = 1$.

For the special case of $A = 1$ and $N = 1$, the solutions are closed form expressions (e.g. Arfken et al., 2013, pp. 1059f):

$$\lambda = \int_\Omega q(s)v(s)\,\mathrm{d}s, \quad \phi(x) = dv(x). \tag{75}$$

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